University Physics
Sears Zemansky

Check email in
BYU → norereply

HW → deadline → Pearson online

2,20 → HW
1,40 → ME →
1,40 → final → } Test (15,20)

Course ID

Zoom meeting
write your full name

→ Kinematic → 2 → 3
→ Dynamics → 4 → 5
→ Work and Energy → 6 → 7
→ momentum → 8
→ Rotational motion about fixed motion → 9
→ Angular momentum
→ Static equilibrium → missing

translational motion
linear motion



# Storyline

Physics 1 => Classical Mechanics
kinematics

# Chapter 2:
# Motion in One Dimension

motion
1) translational (linear) ✓ → particle model → point like
2) rotational ✓
3) vibrational



Physics for Scientists and Engineers, 10e
Raymond A. Serway
John W. Jewett, Jr.

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# Position

**position** $x$: location of particle with respect to chosen reference point

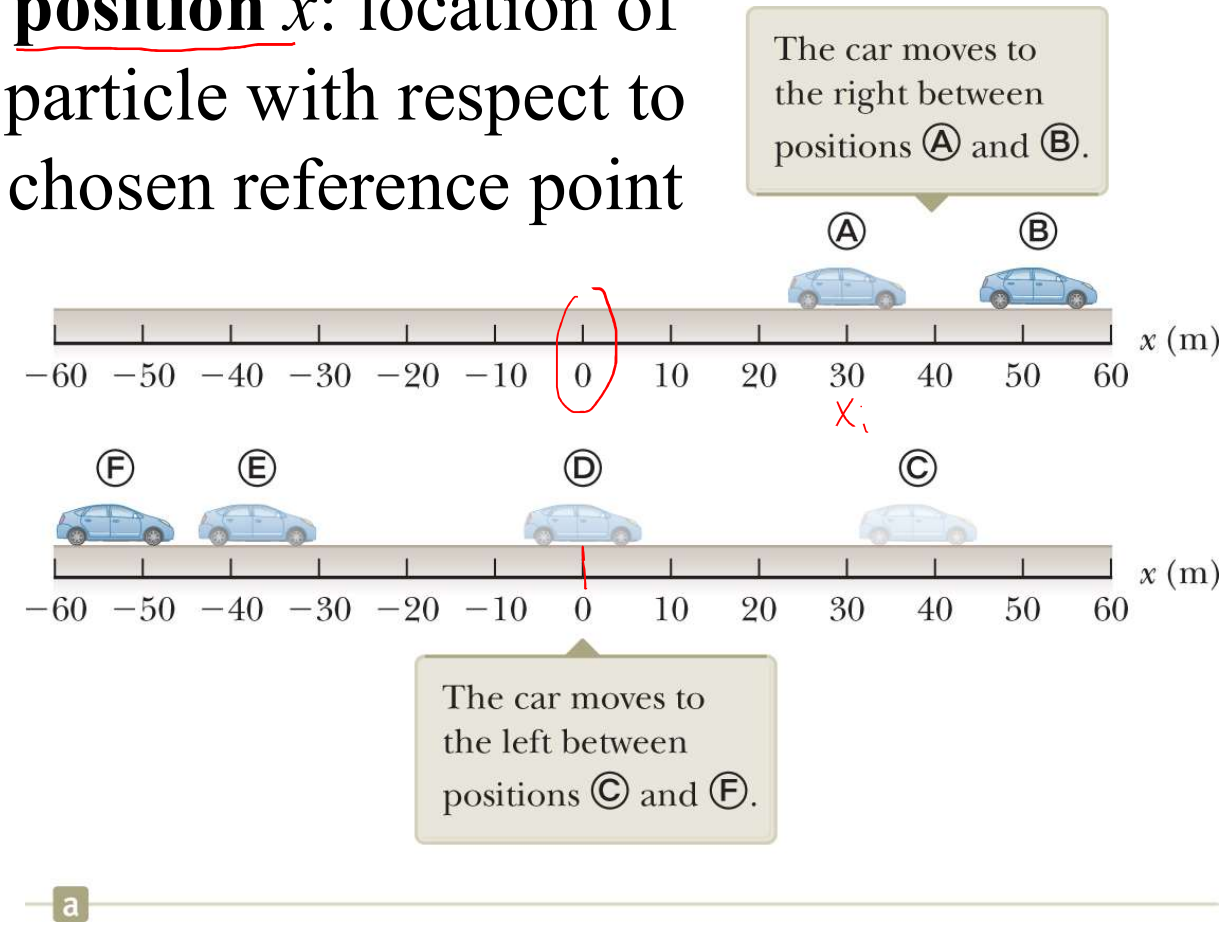
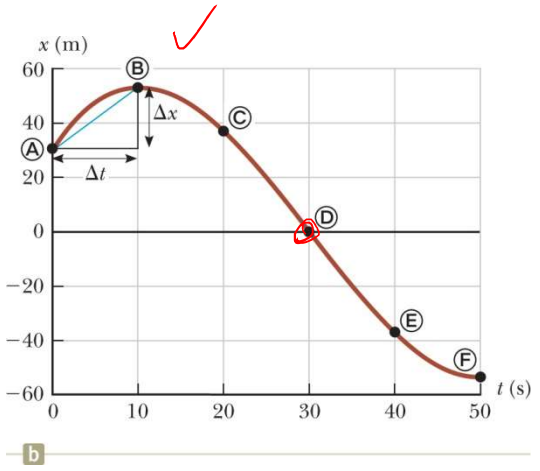



**TABLE 2.1** Position of the Car at Various Times

| Position | $t$ (s) | $x$ (m) |
|---|---|---|
| Ⓐ | 0 | 30 |
| Ⓑ | 10 | 52 |
| Ⓒ | 20 | 38 |
| Ⓓ | 30 | 0 |
| Ⓔ | 40 | −37 |
| Ⓕ | 50 | −53 |



# Alternative Representations

- Pictorial representation
- Tabular representation
- Graphical representation

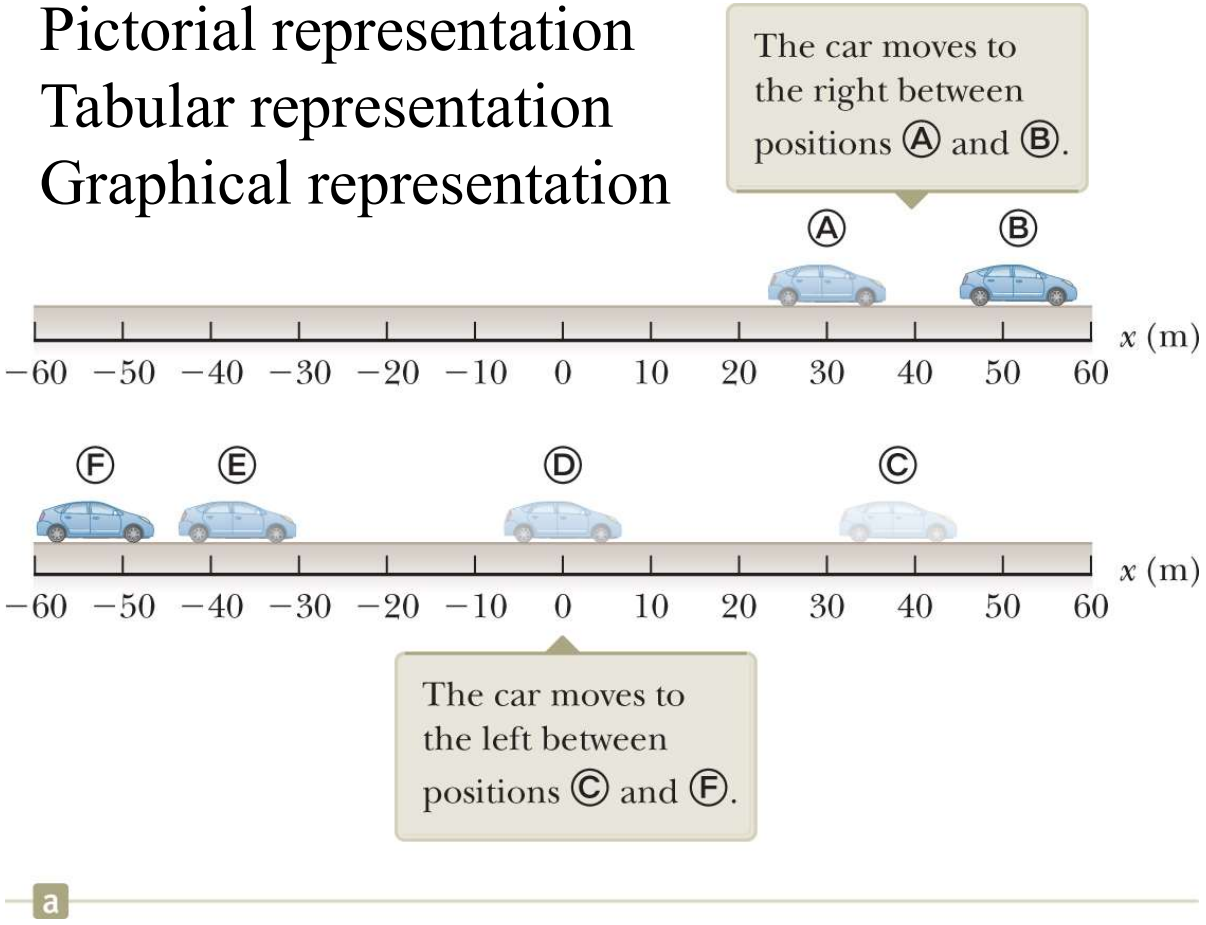
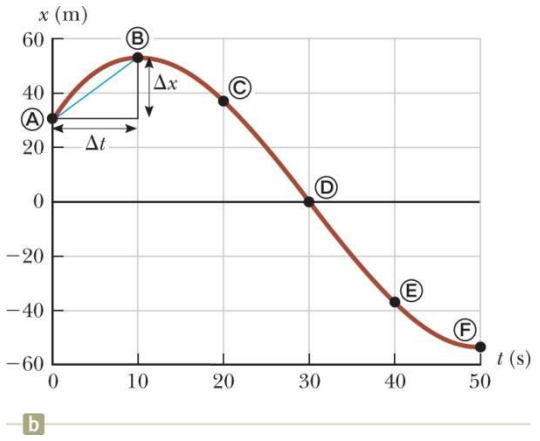



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# Displacement

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**Displacement** $\Delta x$ of particle: change in position in a given time interval

$$\Delta x \equiv x_f - x_i$$

$\Delta x < 0$

$> 0$

$= 0$

# Distance and Displacement



**Vector quantity** requires specification of both direction and magnitude
**Scalar quantity** has numerical value and no direction

# Position, Velocity, and Speed of a Particle

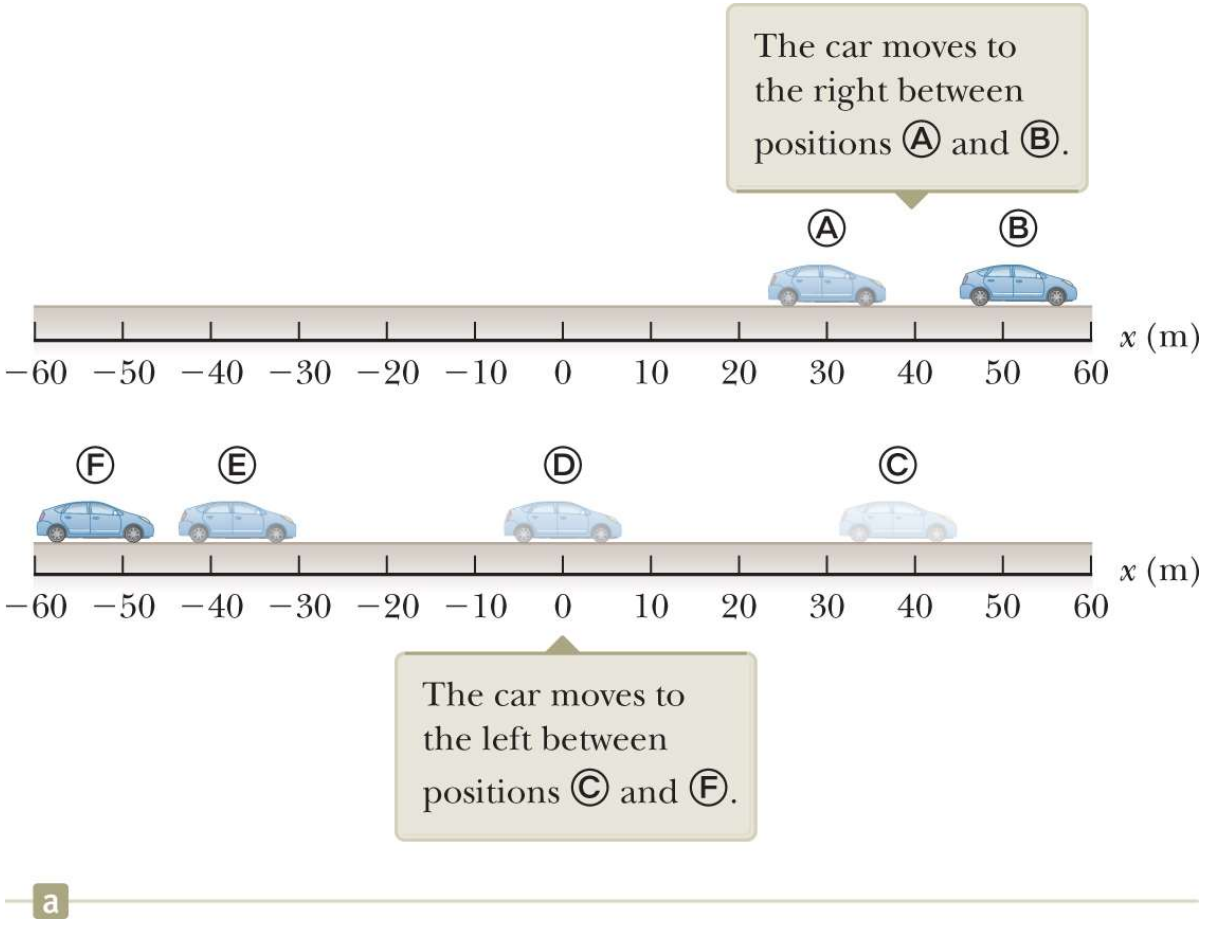
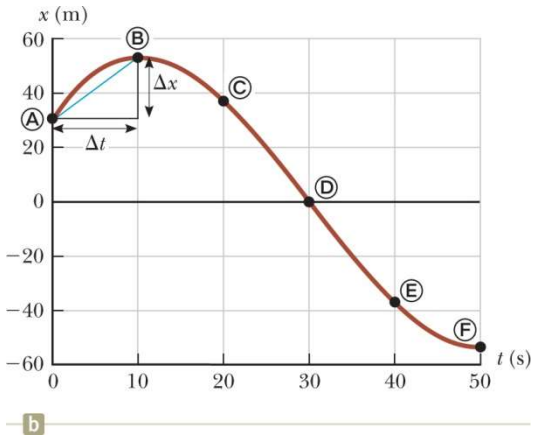



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# Quick Quiz 2.1

Which of the following choices best describes what can be determined *exactly* from the table and figure for the entire 50-s interval?

(a) The distance the car moved.

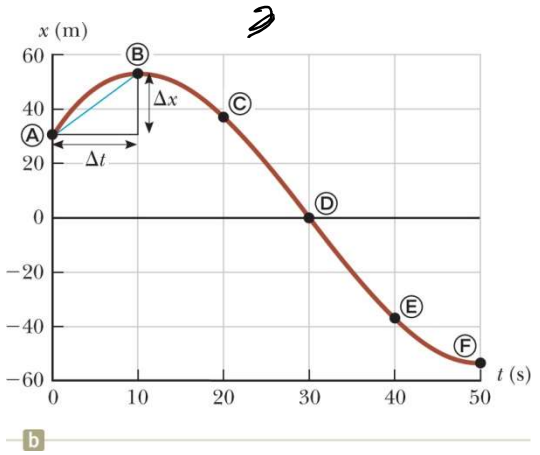
(b) The displacement of the car.

(c) Both (a) and (b).

(d) Neither (a) nor (b).

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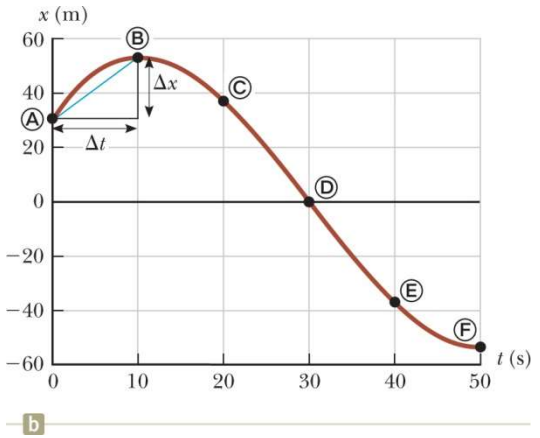
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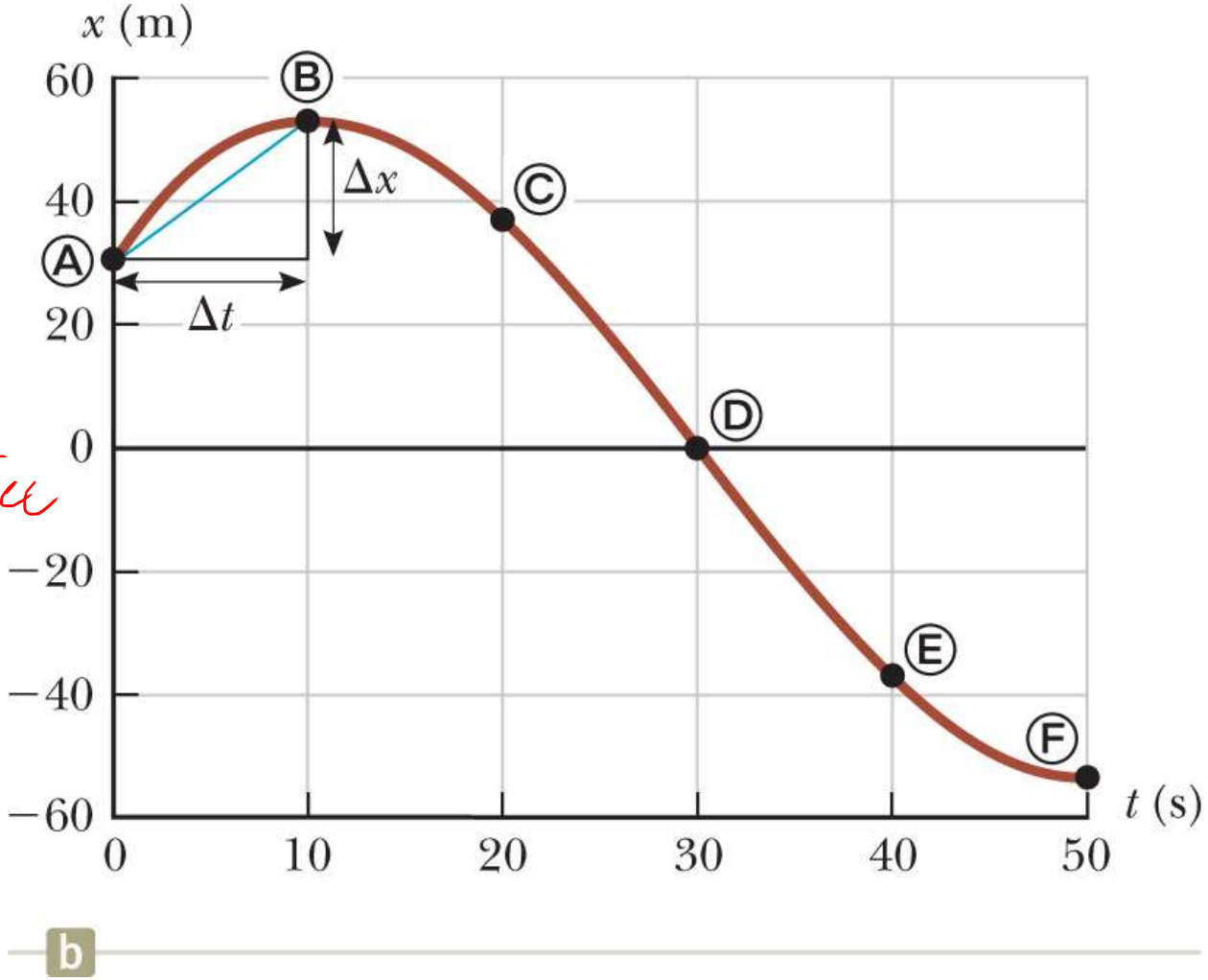


# Position, Velocity, and Speed of a Particle

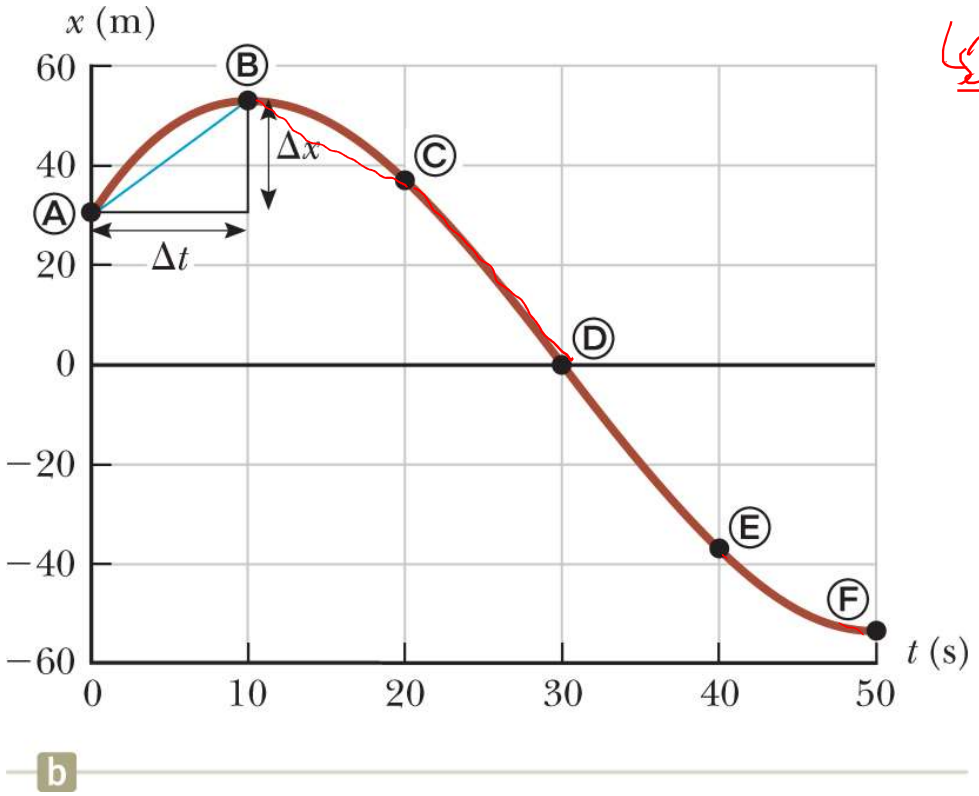
$$v_{x,\text{avg}} \equiv \frac{\Delta x}{\Delta t}$$

m/sec

Average velocity



# Average Velocity




Example: $\dfrac{52\text{ m} - 30\text{ m}}{10\text{ s} - 0} = 2.2\text{ m/s}$

# Average Speed

**Average speed** of particle (scalar quantity): total distance $d$ traveled divided by elapsed time $\Delta t$



$$v_{\text{avg}} \equiv \frac{d}{\Delta t}$$

$$v_{\text{avg}} = \frac{+75 \text{ m}}{55.0 \text{ s}} = +1.36 \text{ m/s}$$

$$\text{average speed} = \frac{125 \text{ m}}{55.0 \text{ s}} = 2.27 \text{ m/s}$$

# Quick Quiz 2.2

Under which of the following conditions is the magnitude of the average velocity of a particle moving in one dimension smaller than the average speed over the time interval?

(a) A particle moves in the $+x$ direction without reversing.

(b) A particle moves in the $-x$ direction without reversing.

(c) A particle moves in the $+x$ direction and then reverses the direction of its motion.

(d) There are no conditions for which this is true.

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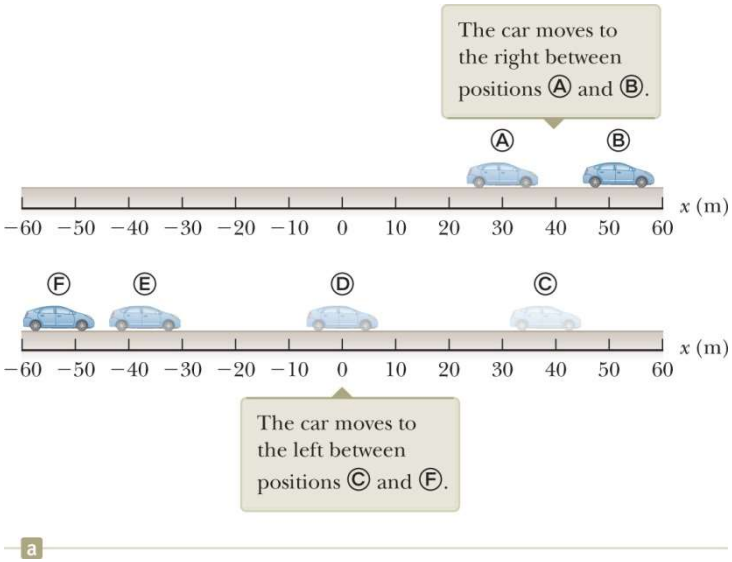
(b) A particle moves in the $-x$ direction without reversing.

**(c) A particle moves in the $+x$ direction and then reverses the direction of its motion.**

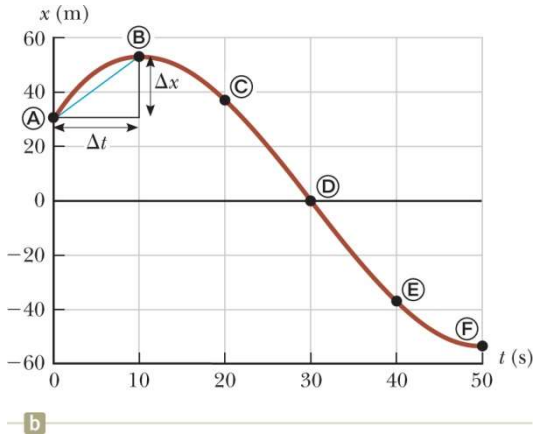
(d) There are no conditions for which this is true.

# Example 2.1: Calculating the Average Velocity and Speed

Find the displacement, average velocity, and average speed of the car in the figure between positions A and F.




$$\Delta x = x_F - x_A$$
$$= -53\ \text{m} - 30\ \text{m}$$
$$= -83\ \text{m}$$



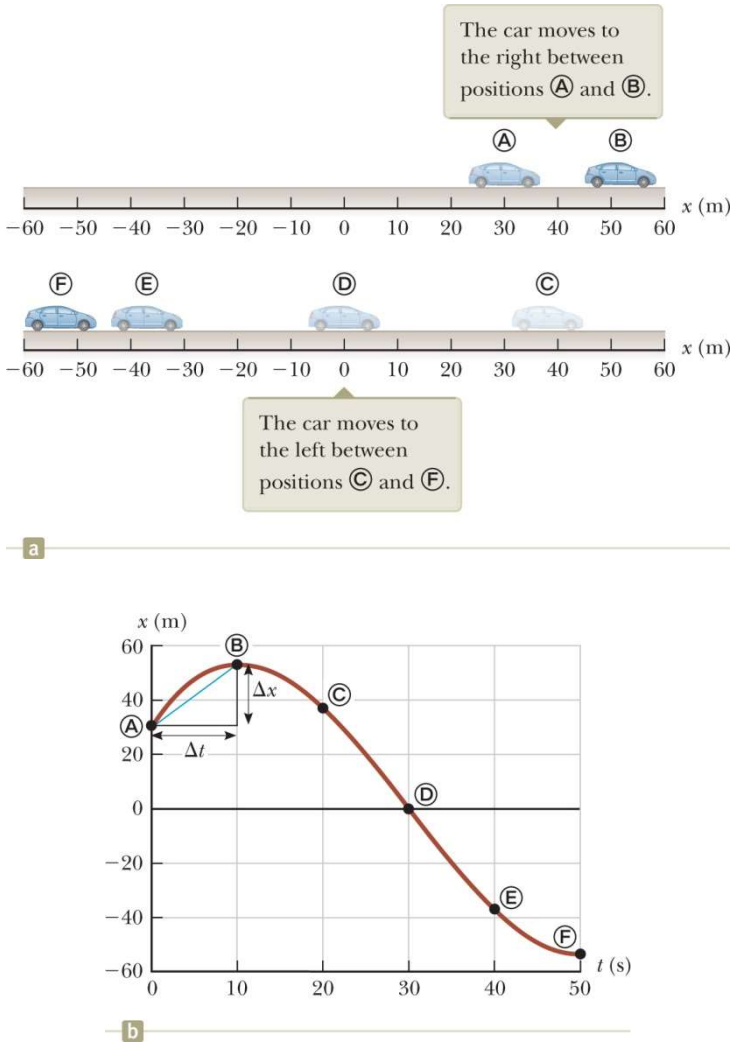
# Example 2.1: Calculating the Average Velocity and Speed

Average velocity:

$$v_{x,\text{avg}} = \frac{x_\text{F} - x_\text{A}}{t_\text{F} - t_\text{A}}$$

$$= \frac{-53\ \text{m} - 30\ \text{m}}{50\ \text{s} - 0\ \text{s}}$$

$$= \frac{-83\ \text{m}}{50\ \text{s}} = \boxed{-1.7\ \text{m/s}}$$



# Example 2.1: Calculating the Average Velocity and Speed

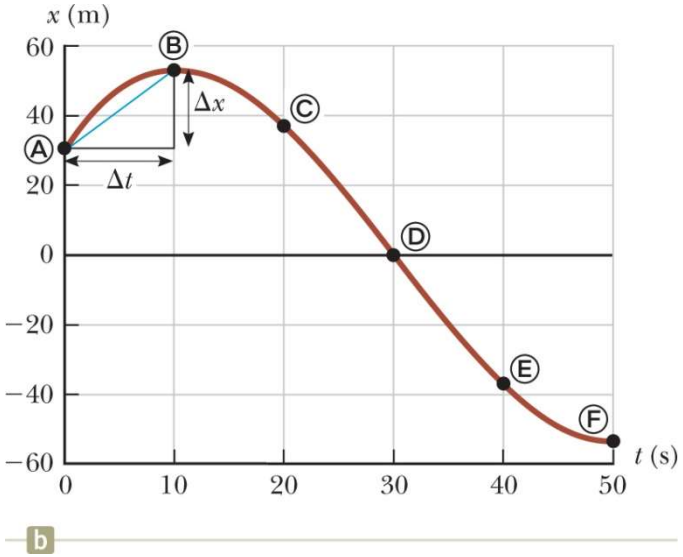
Average speed:

$$v_{\text{avg}} = \frac{127 \text{ m}}{50.0 \text{ s}} = \boxed{2.54 \text{ m/s}}$$

**TABLE 2.1** Position of the Car at Various Times

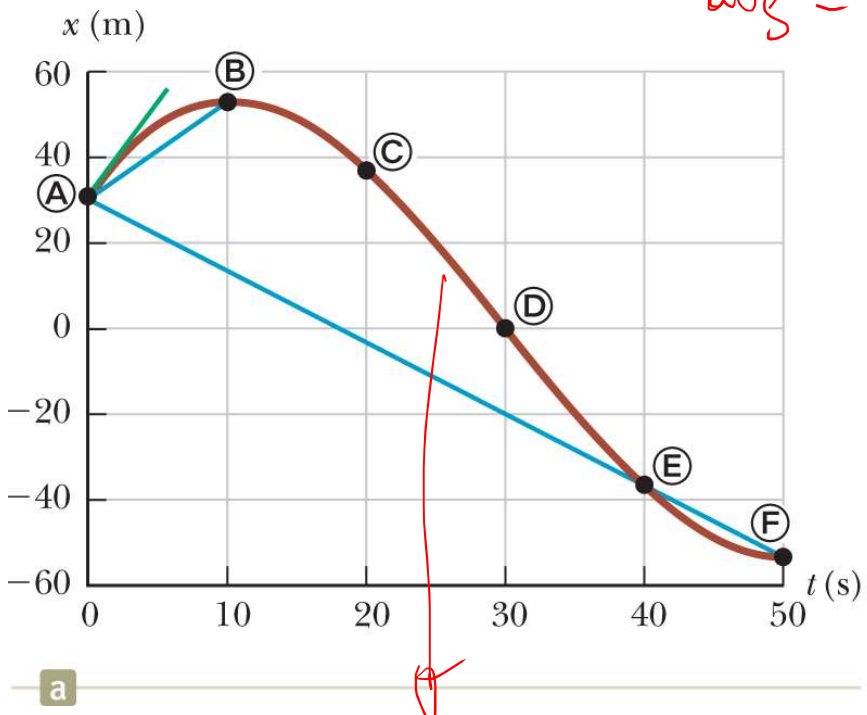
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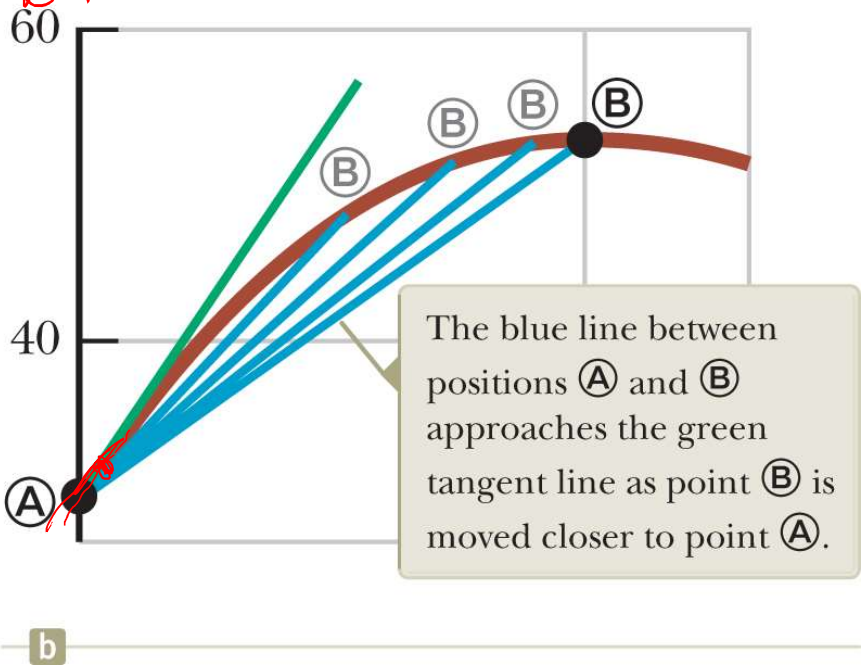
22
14
38
37
16
127m



# Instantaneous Velocity and Speed

$v_{avg} = \frac{\Delta x}{\Delta t}$




$x(t) = 5t + 3t^2$

$v_x = \frac{dx}{dt} = 5 + 6t$

$$v_x \equiv \lim_{\Delta t \to 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt} \rightarrow$$

$v_x = \frac{dx}{dt}$

# Quick Quiz 2.3

Members of the highway patrol are more interested in

(a) your average speed or

(b) your instantaneous speed

as you drive.

# Quick Quiz 2.3

Members of the highway patrol are more interested in

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# Conceptual Example 2.2: The Velocity of Different Objects

Consider the one-dimensional motion of a ball thrown directly upward, which rises to a highest point and falls back into the thrower's hand. Are there any points in the motion of this object at which the instantaneous velocity has the same value as the average velocity over the entire motion? If so, identify the point(s).



**One point at top of motion**

# Conceptual Example 2.2: The Velocity of Different Objects

Consider the one-dimensional motion of a race car which starts from rest and speeds up to 100 m/s. Are there any points in the motion of this object at which the instantaneous velocity has the same value as the average velocity over the entire motion?

**One instant somewhere between 0 and 100 m/s**

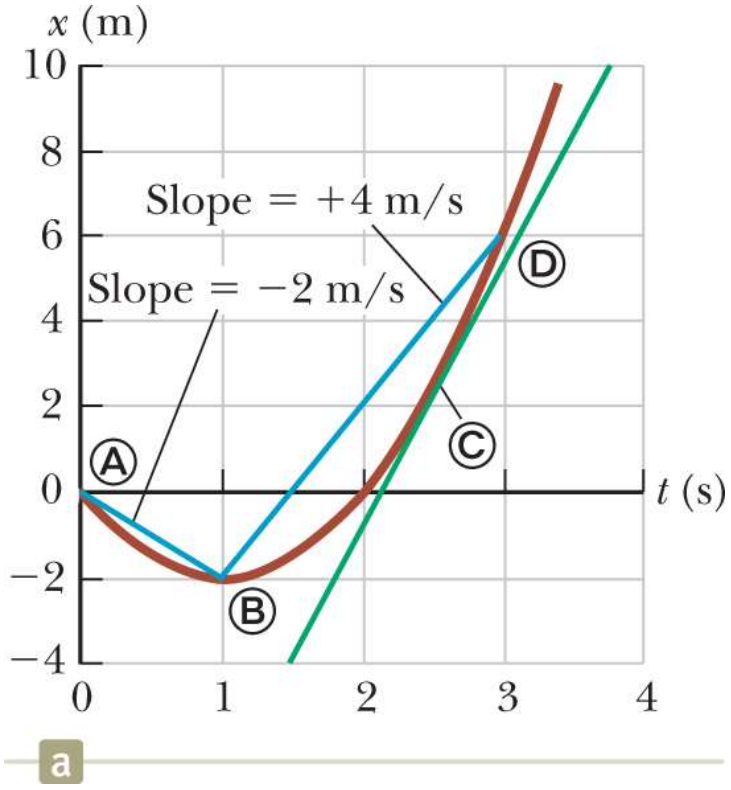
# Conceptual Example 2.2: The Velocity of Different Objects

Consider the one-dimensional motion of a spacecraft that drifts through space at constant velocity. Are there any points in the motion of this object at which the instantaneous velocity has the same value as the average velocity over the entire motion? If so, identify the point(s).

**for ALL positions of the spacecraft**

# Example 2.3: Average and Instantaneous Velocity

A particle moves along the $x$ axis. Its position varies with time according to the expression $x = -4t + 2t^2$, where $x$ is in meters and $t$ is in seconds. The position–time graph for this motion is shown in the figure. Because the position of the particle is given by a mathematical function, the motion of the particle is known at all times. Notice that the particle moves in the negative $x$ direction for the first second of motion, is momentarily at rest at the moment $t = 1$ s, and moves in the positive $x$ direction at times $t > 1$ s.



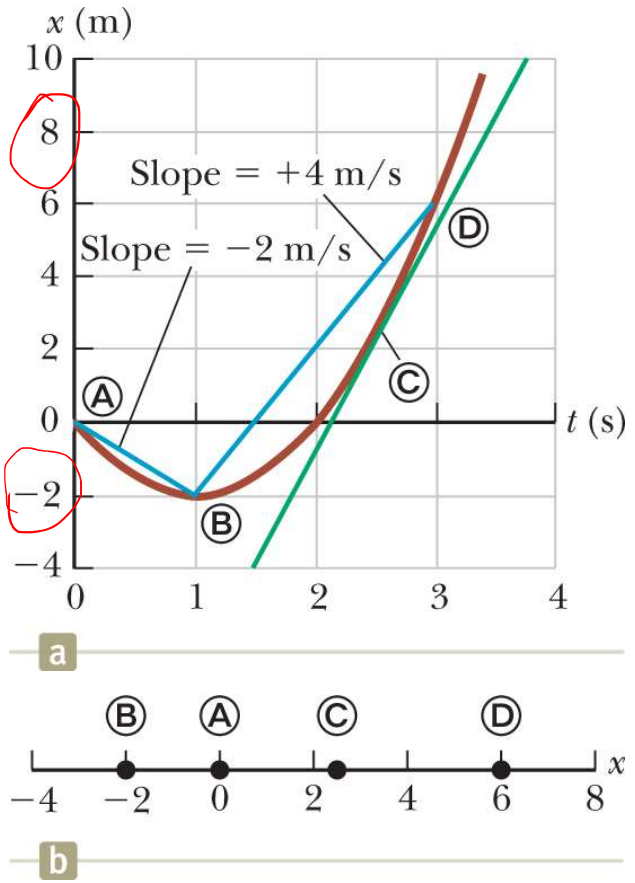
# Example 2.3: Average and Instantaneous Velocity

(A) Determine the displacement of the particle in the time intervals $t = 0$ to $t = 1$ s and $t = 1$ s to $t = 3$ s.

$x(t) = -4t + 2t^2$

$$\Delta x_{A\to B} = x_f - x_i = x_B - x_A$$
$$= \left[-4(1) + 2(1)^2\right] - \left[-4(0) + 2(0)^2\right]$$
$$= \boxed{-2 \text{ m}}$$

$$\Delta x_{B\to D} = x_f - x_i = x_D - x_B$$
$$= \left[-4(3) + 2(3)^2\right] - \left[-4(1) + 2(1)^2\right]$$
$$= \boxed{+8 \text{ m}}$$

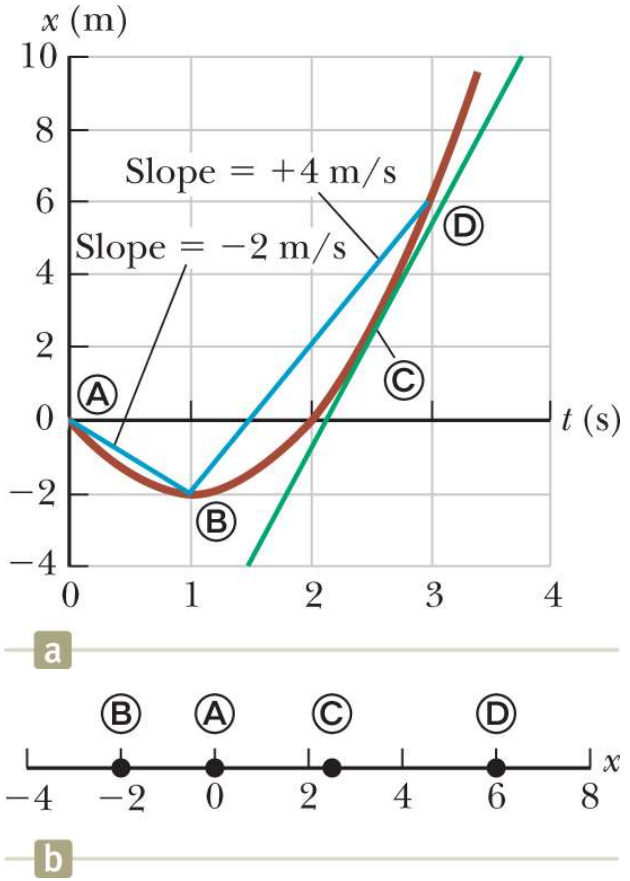


# Example 2.3: Average and Instantaneous Velocity

(B) Calculate the average velocity during these two time intervals.

$$v_{x,\text{avg}(A\to B)} = \frac{\Delta x_{A\to B}}{\Delta t} = \frac{-2\text{ m}}{1\text{ s}} = \boxed{-2\text{ m/s}}$$

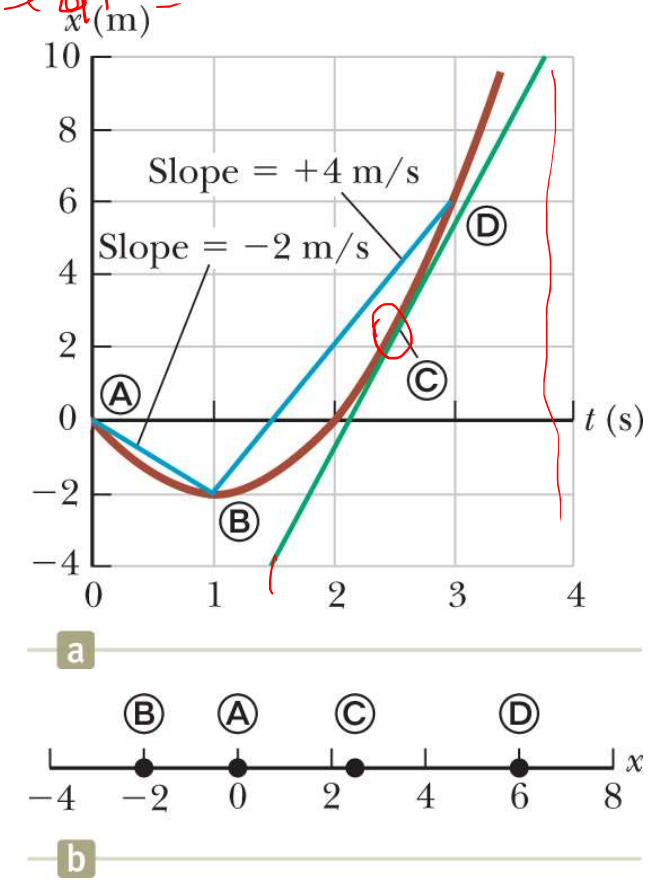
$$v_{x,\text{avg}(B\to D)} = \frac{\Delta x_{B\to D}}{\Delta t} = \frac{8\text{ m}}{2\text{ s}} = \boxed{+4\text{ m/s}}$$



# Example 2.3:
# Average and Instantaneous Velocity

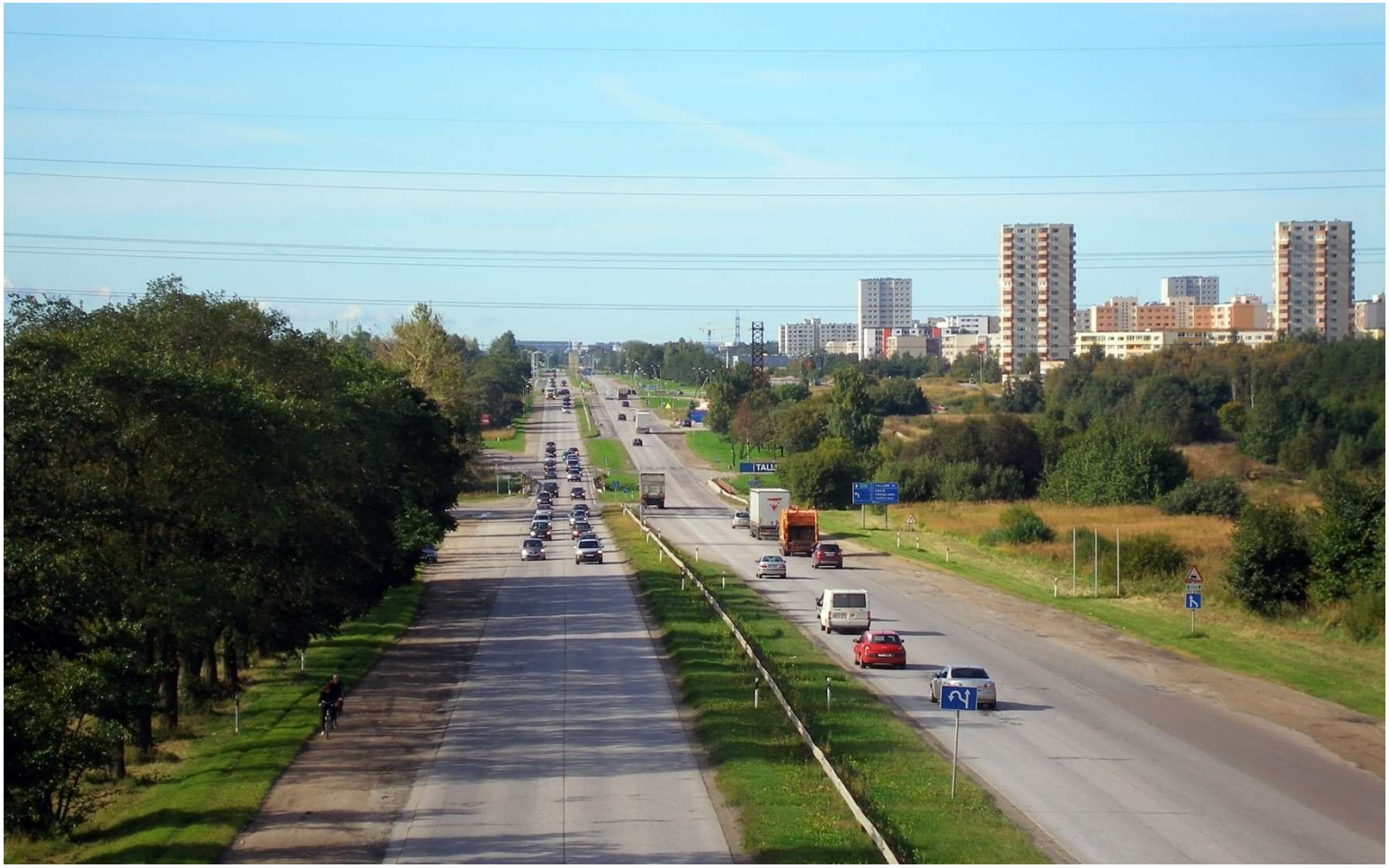
(C) Find the instantaneous velocity of the particle at $t = 2.5$ s.

$$v_x = \frac{10\text{ m} - (-4\text{ m})}{3.8\text{ s} - 1.5\text{ s}} = \boxed{+6\text{ m/s}}$$



# Analysis Model: Particle Under Constant Velocity

**Analysis model**: represents common situation when solving physics problems



# Analysis Model: Particle Under Constant Velocity

$$v_{x,\text{avg}} \equiv \frac{\Delta x}{\Delta t} \rightarrow v_x = \frac{\Delta x}{\Delta t}$$

$v_x = v_{x,\text{avg}}$

$$\Delta x = x_f - x_i \rightarrow v_x = \frac{x_f - x_i}{\Delta t}$$

$$x_f = x_i + v_x \Delta t$$ → for constant $v_x$

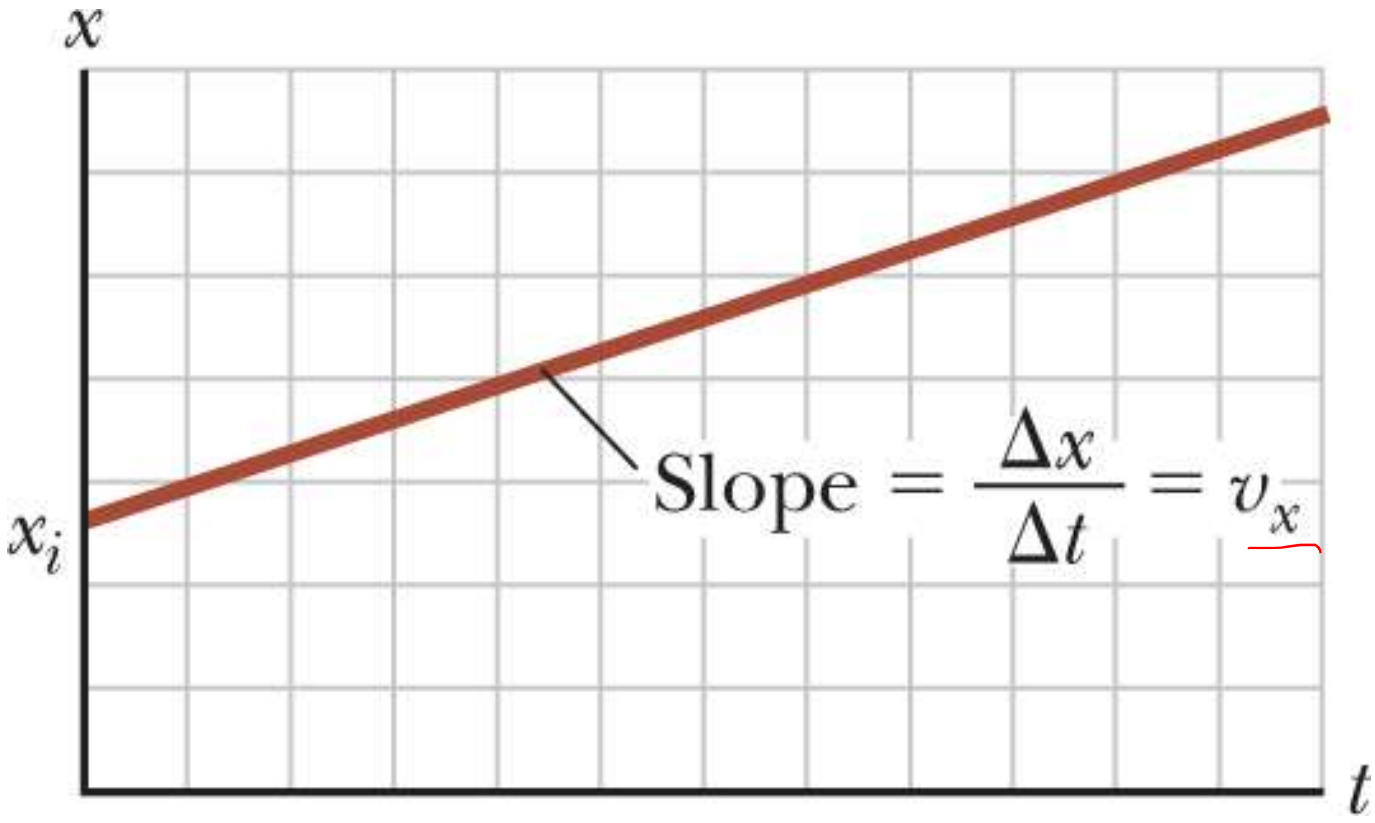
$$\boxed{x_f = x_i + v_x t \quad (\text{for constant } v_x)}$$

# Analysis Model: Particle Under Constant Velocity

$$v_x = \frac{\Delta x}{\Delta t}$$

$$x_f = x_i + v_x t \quad \left(\text{for constant } v_x\right)$$

# Analysis Model: Particle Under Constant Velocity




$$x = x_i + v_i t$$

# Example 2.4:
# Modeling a Runner as a Particle

A kinesiologist is studying the biomechanics of the human body. (*Kinesiology* is the study of the movement of the human body. Notice the connection to the word *kinematics*.) She determines the velocity of an experimental subject while he runs along a straight line at a constant rate. The kinesiologist starts the stopwatch at the moment the runner passes a given point and stops it after the runner has passed another point 20 m away. The time interval indicated on the stopwatch is 4.0 s.

# Example 2.4:
# Modeling a Runner as a Particle

(A) What is the runner's velocity?

$$v_x = \frac{\Delta x}{\Delta t} = \frac{x_f - x_i}{\Delta t} = \frac{20\ \text{m} - 0}{4.0\ \text{s}} = 5.0\ \text{m/s}$$

# Example 2.4:
# Modeling a Runner as a Particle

(B) If the runner continues his motion after the stopwatch is stopped, what is his position after 10 s have passed?

$$x_f = x_i + v_x t = 0 = 0 + (5.0\ \text{m/s})(10\ \text{s}) = \boxed{50\ \text{m/s}}$$

# Analysis Model: Particle Under Constant Speed

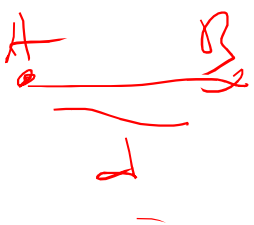
$l = r\theta = 2\pi r$




if a person moves with constant speed along a circular path.

$$v = \frac{d}{\Delta t}$$



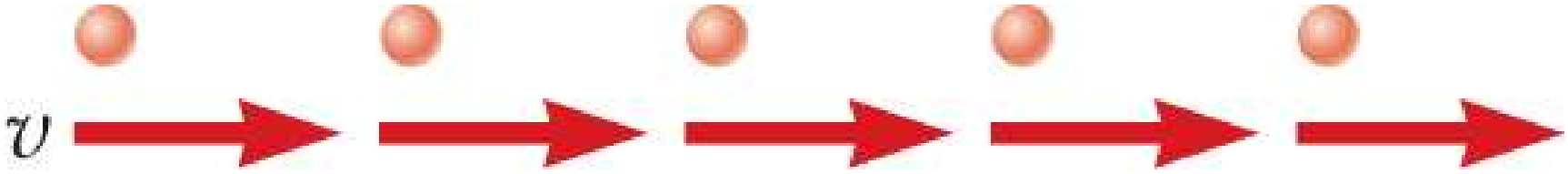

$$v = \frac{d}{\Delta t} \rightarrow \Delta t = \frac{d}{v} = \frac{2\pi r}{v} = \frac{2\pi(10.0\text{ m})}{5.00\text{ m/s}} = 12.6\text{ s}$$

time required for completing one full trip.

# Analysis Model: Particle Under Constant Velocity

constant velocity: $v_x = \dfrac{\Delta x}{\Delta t}$

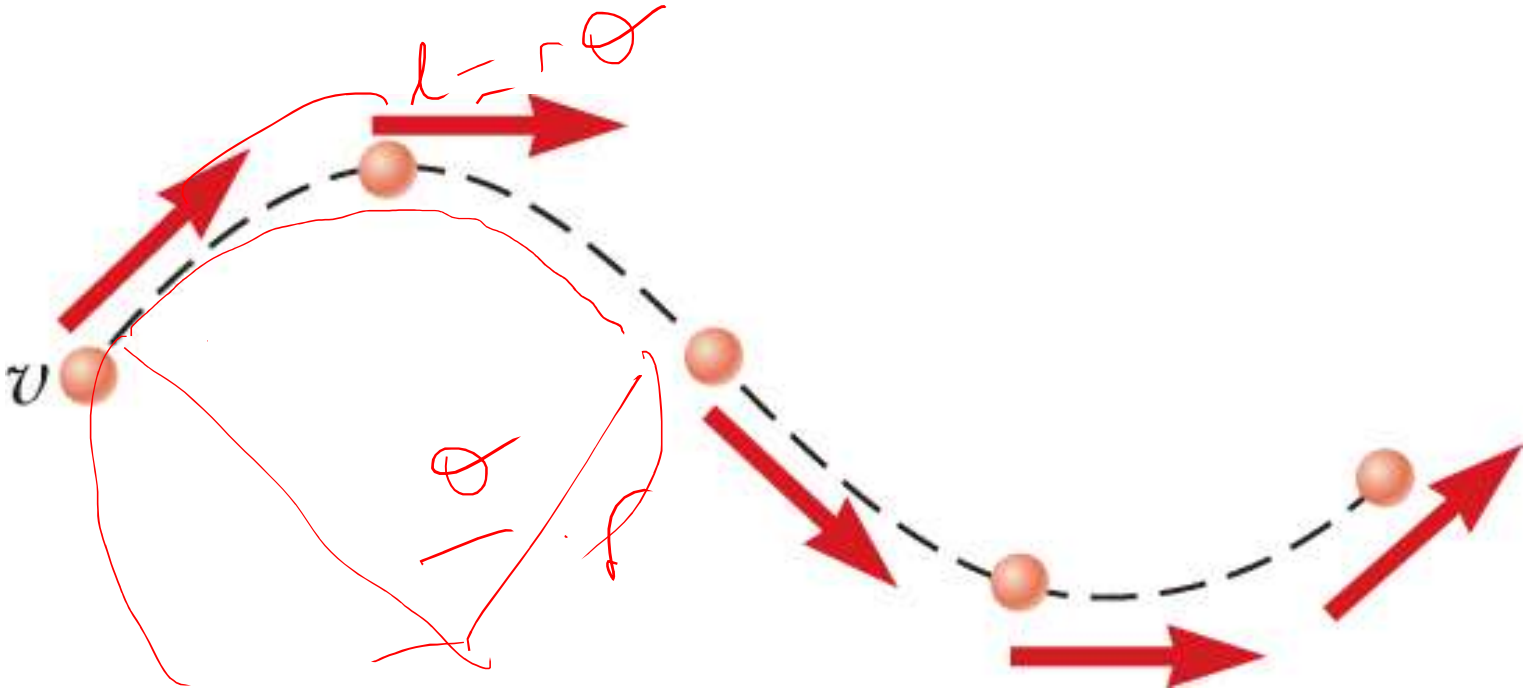
position as a function of time: $x_f = x_i + v_x t$



# Analysis Model: Particle Under Constant Speed

$|v|$ ⇒ magnitude → speed

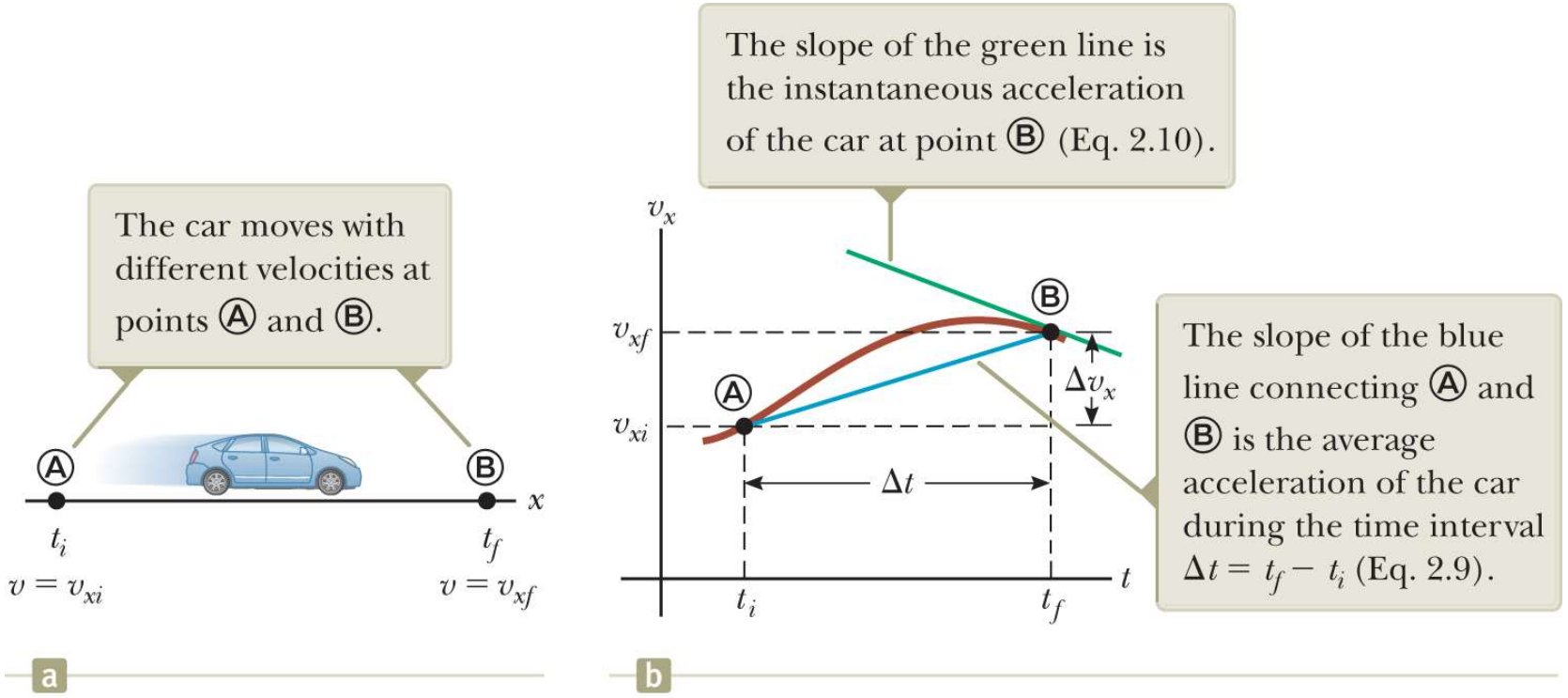
constant speed: $v = \dfrac{d}{\Delta t}$



# Analysis Model Approach to Problem-Solving

- Conceptualize
- Categorize
- Analyze
- Finalize

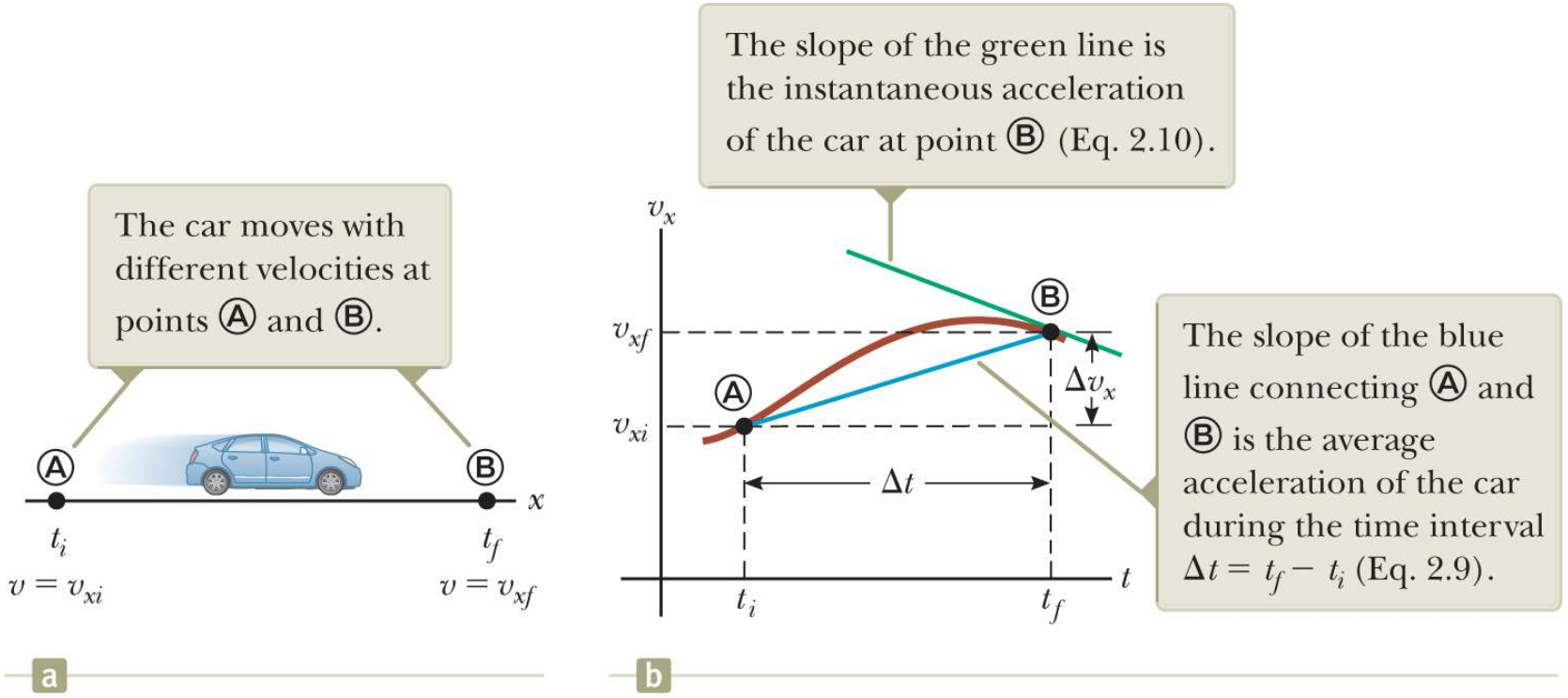
# Average Acceleration




$$a_{x,\text{avg}} \equiv \frac{\Delta v_x}{\Delta t} = \frac{v_{xf} - v_{xi}}{t_f - t_i}$$

$\frac{M}{Sec^2}$ $>0$ $=0$ $<0$

# Instantaneous Acceleration

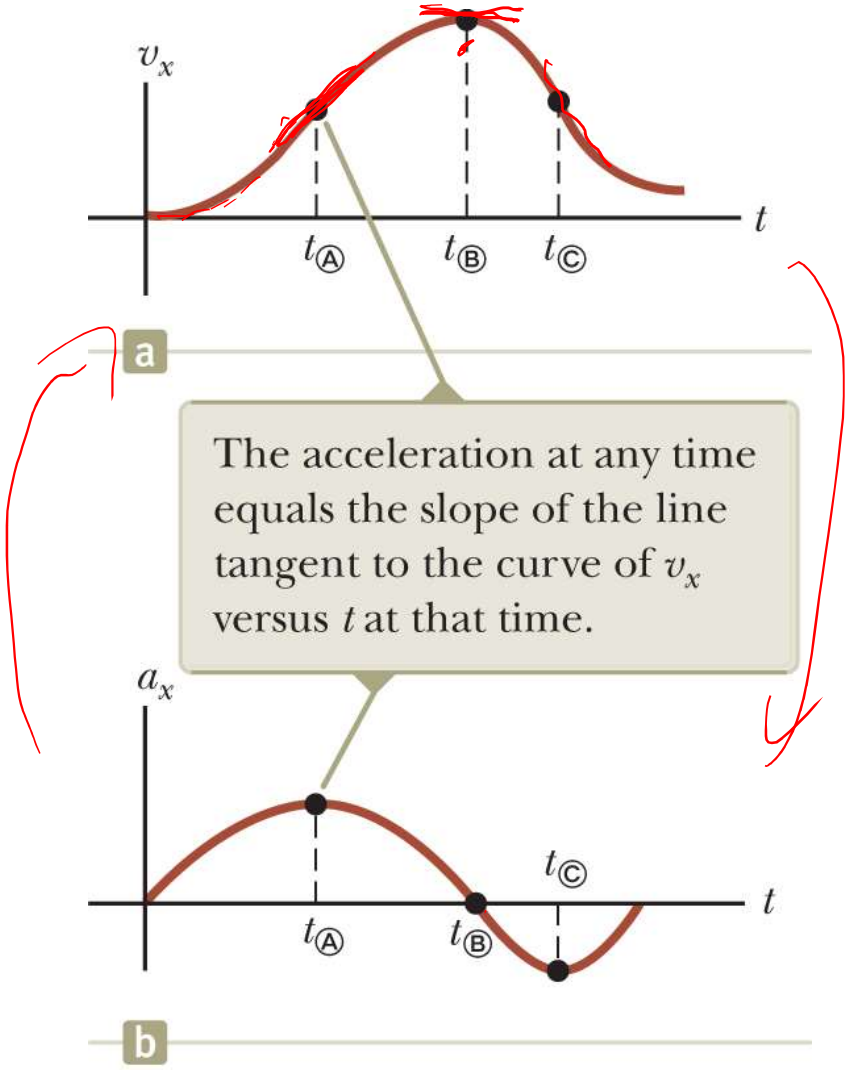



$V(t) = at^2 + bt$

$\frac{dv}{dt} = 2at + b$

$$a_x \equiv \lim_{\Delta t \to 0} \frac{\Delta v_x}{\Delta t} = \frac{dv_x}{dt}$$

# Acceleration vs. Time Graph

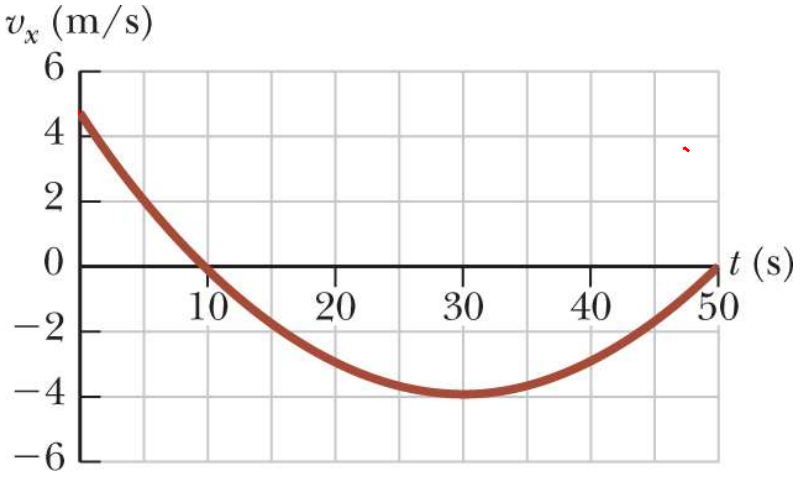


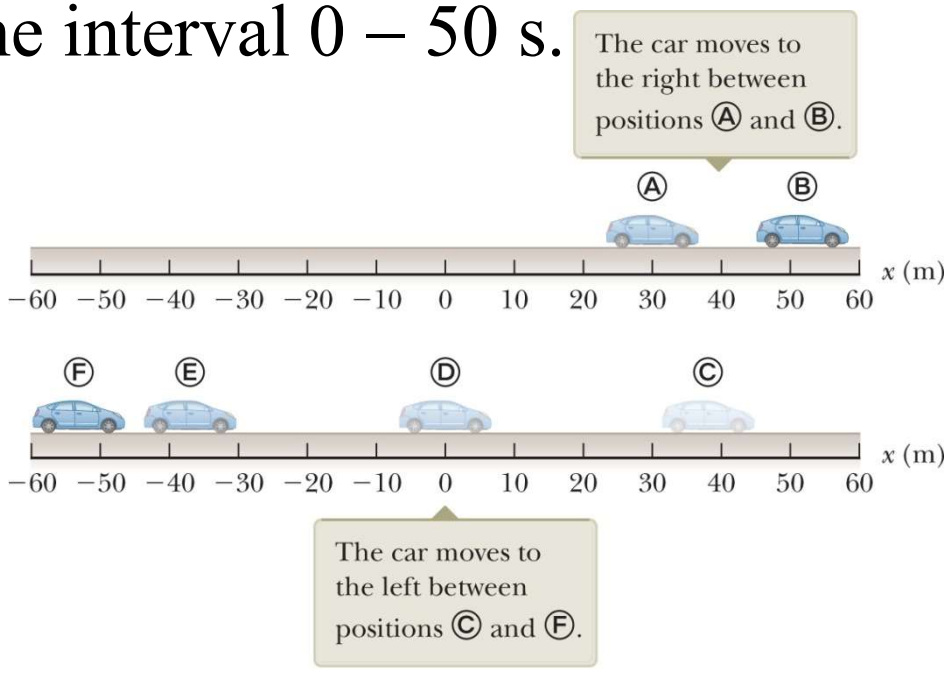
# Quick Quiz 2.4

Make a velocity–time graph for the car in the figure. Suppose the speed limit for the road on which the car is driving is 30 km/h.

True or False? The car exceeds the speed limit at some time within the time interval 0 – 50 s.

**False**



# Force and Acceleration

The force on an object is proportional to the acceleration of the object.

$$F_x \propto a_x$$

# Quick Quiz 2.5

If a car is traveling eastward and slowing down, what is the direction of the force on the car that causes it to slow down?

(a) eastward

(b) westward

(c) neither eastward nor westward

# Quick Quiz 2.5

If a car is traveling eastward and slowing down, what is the direction of the force on the car that causes it to slow down?

(a) eastward

**(b) westward**

(c) neither eastward nor westward

# Acceleration

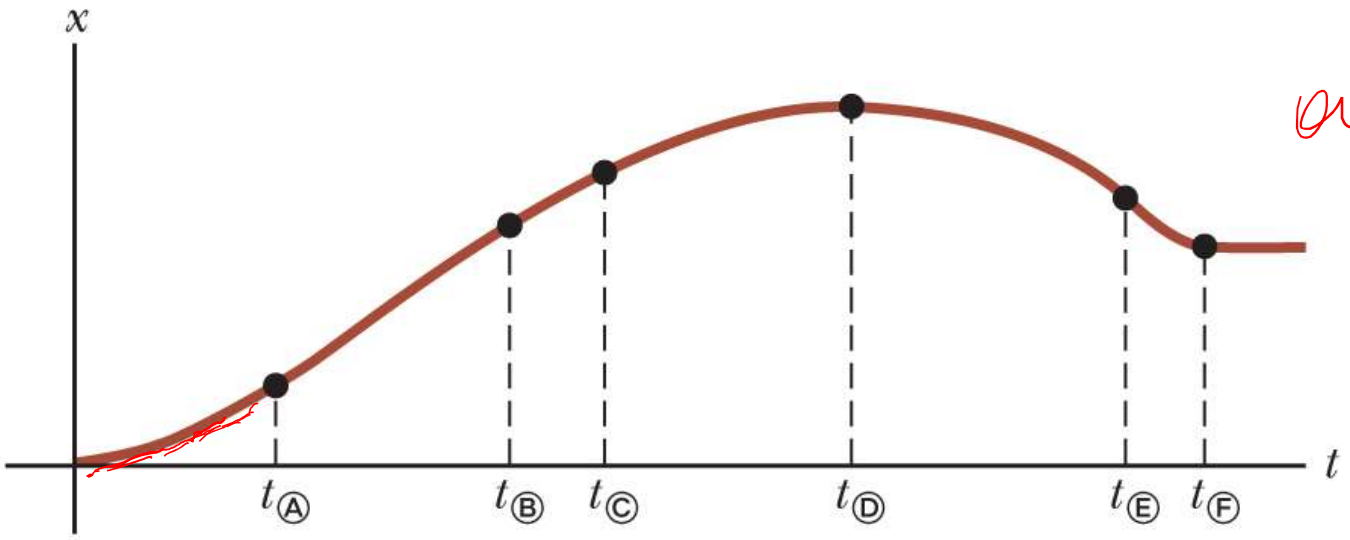
$$a_x = \frac{dv_x}{dt} = \frac{d}{dt}\left(\frac{dx}{dt}\right) = \frac{d^2x}{dt^2}$$

# Conceptual Example 2.5: Graphical Relationships Between $x$, $v_x$, and $a_x$

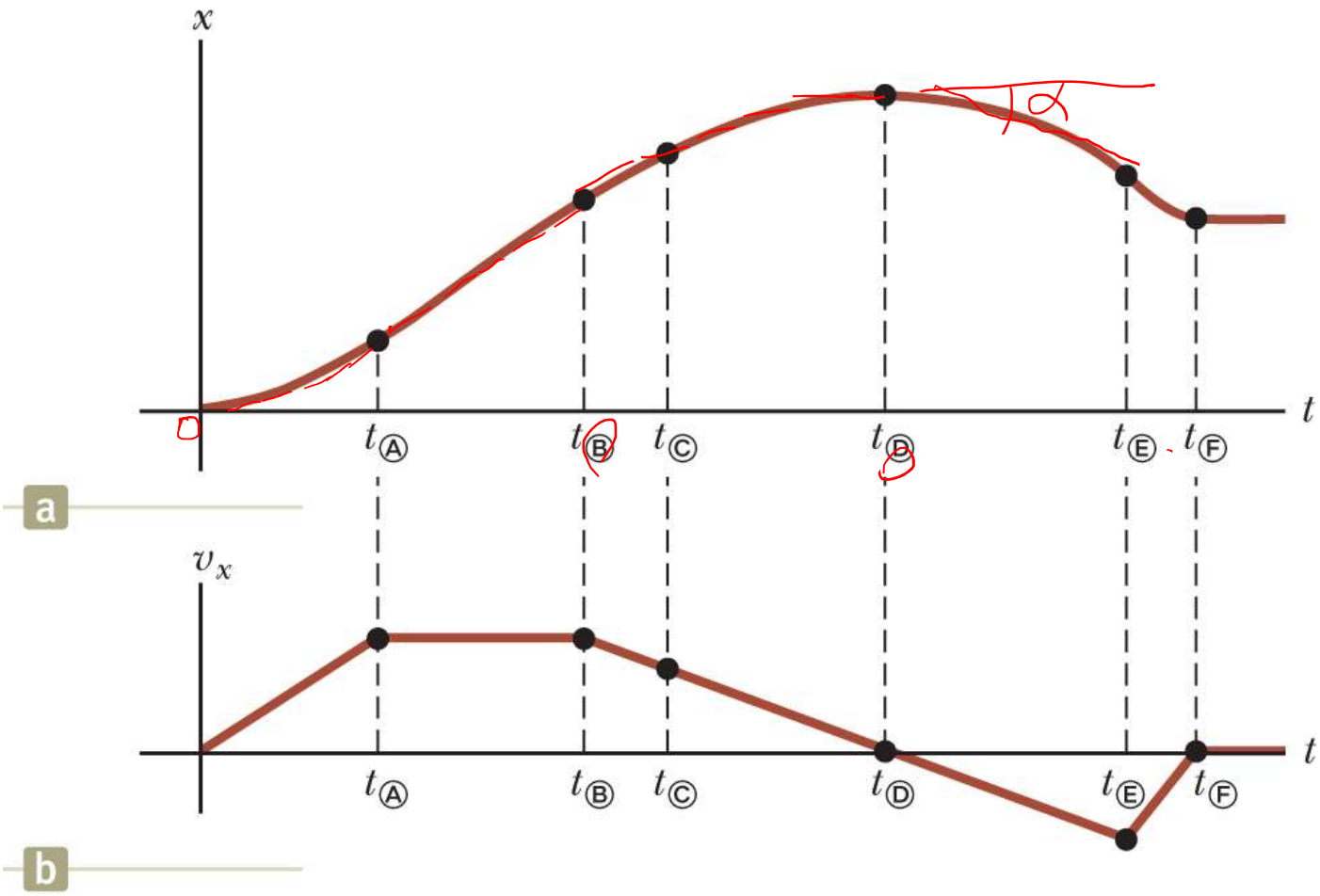
The position of an object moving along the $x$ axis varies with time as in the figure. Graph the velocity versus time and the acceleration versus time for the object.

$v = \frac{dx}{dt}$

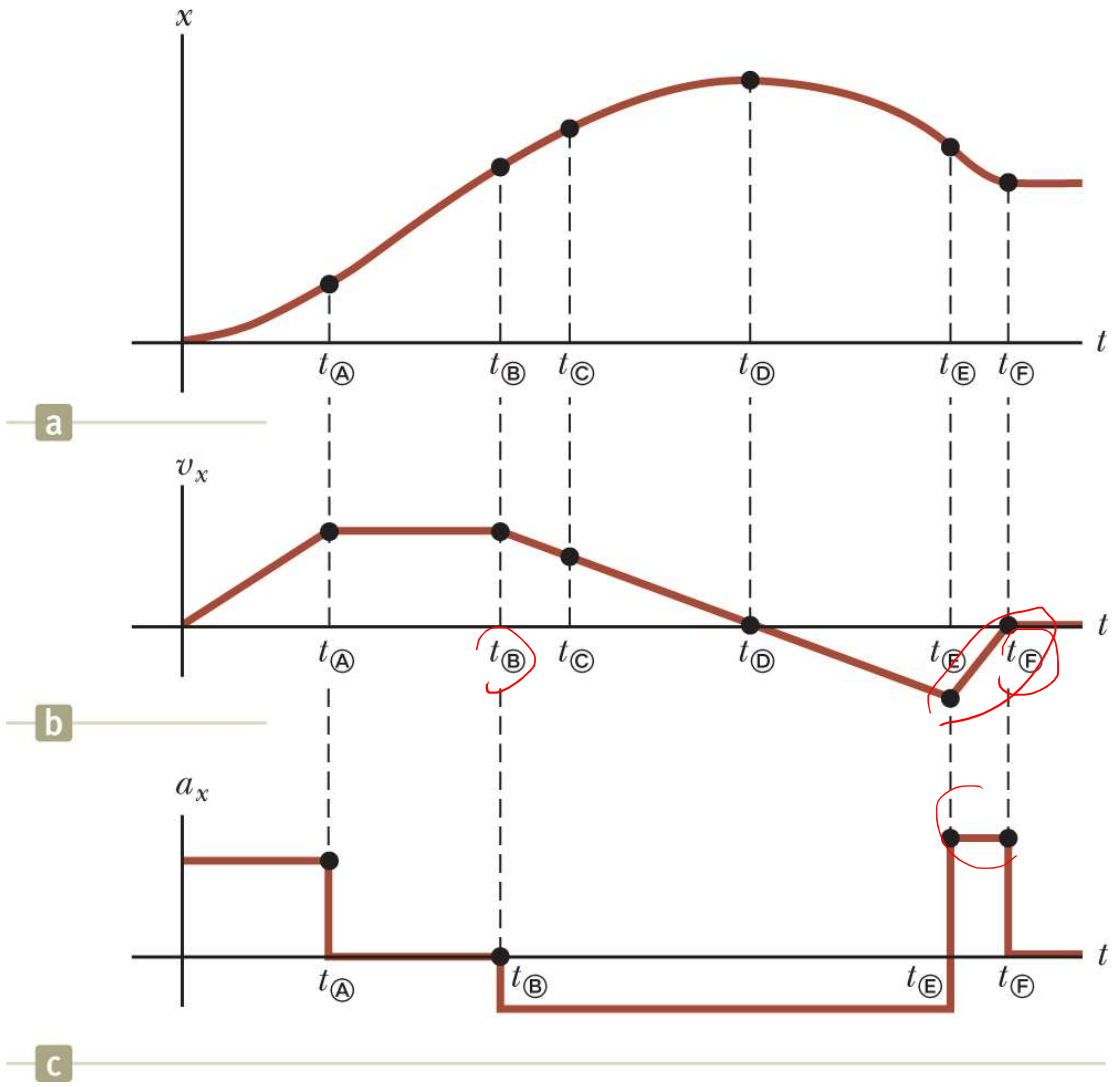
$a = \frac{dv}{dt}$



# Conceptual Example 2.5: Graphical Relationships Between $x$, $v_x$, and $a_x$

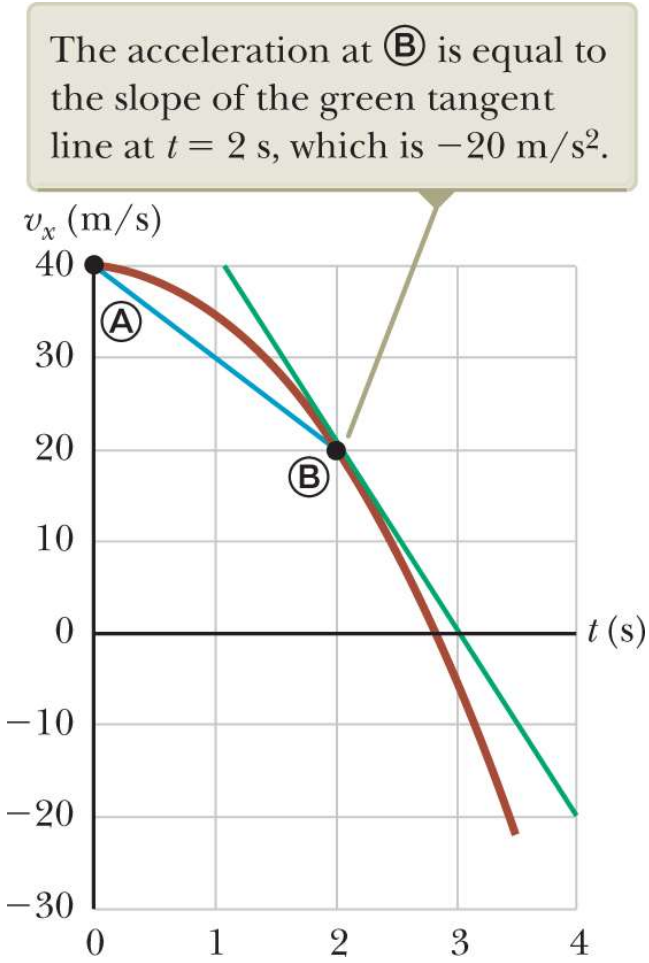


# Conceptual Example 2.5: Graphical Relationships Between $x$, $v_x$, and $a_x$



# Example 2.6: Average and Instantaneous Acceleration

The velocity of a particle moving along the $x$ axis varies according to the expression $v_x = 40 - 5t^2$, where $v_x$ is in meters per second and $t$ is in seconds.



# Example 2.6: Average and Instantaneous Acceleration

$v_x = 40 - 5t^2$

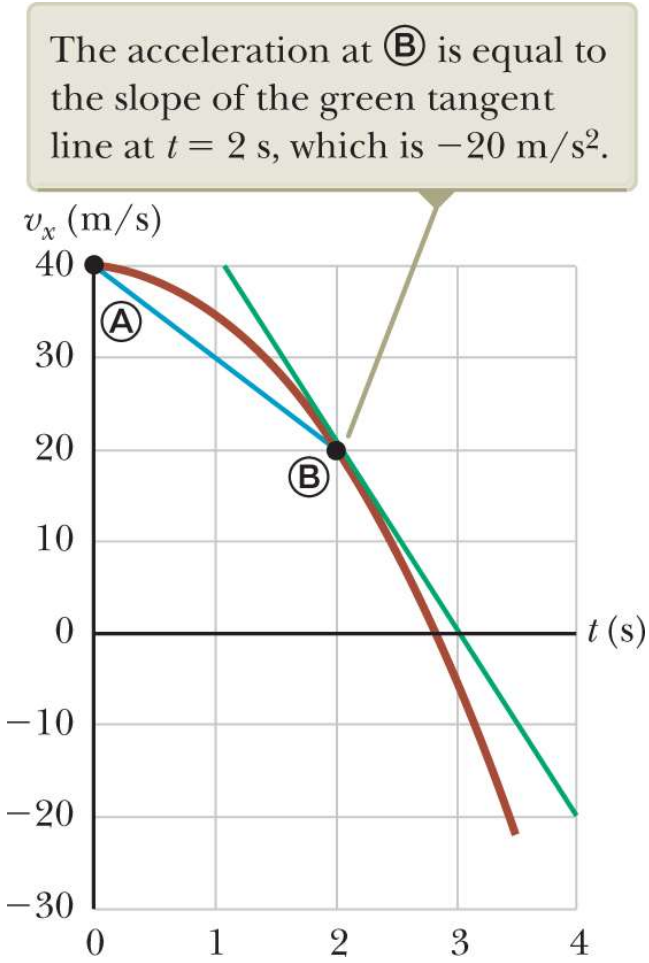
(A) Find the average acceleration in the time interval $t = 0$ to $t = 2.0$ s.

$$v_{xA} = 40 - 5t_A^{\,2} = 40 - 5(0)^2 = +40 \text{ m/s}$$

$$v_{xB} = 40 - 5t_B^{\,2} = 40 - 5(2.0)^2 = +20 \text{ m/s}$$

$$a_{x,\text{avg}} = \frac{v_{xf} - v_{xi}}{t_f - t_i} = \frac{v_{xB} - v_{xA}}{t_B - t_A} = \frac{20 \text{ m/s} - 40 \text{ m/s}}{2.0 \text{ s} - 0 \text{ s}}$$

$$= \boxed{-10 \text{ m/s}^2}$$



# Example 2.6: Average and Instantaneous Acceleration

(B) Determine the acceleration at $t = 2.0$ s.

$v = 40 - 5t^2$

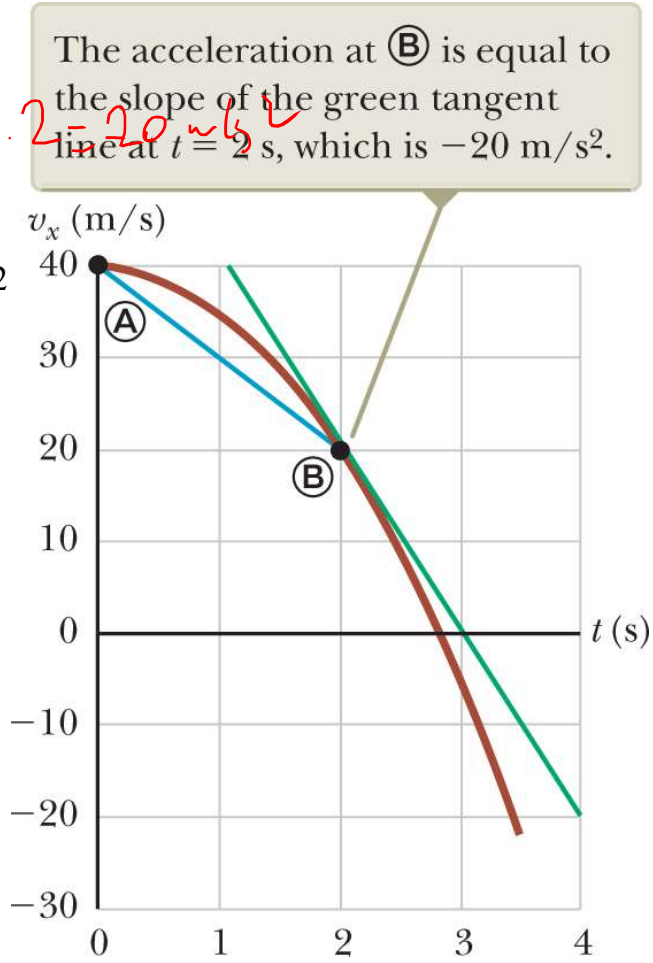
$\frac{dv}{dt} = -10t = -10 \cdot 2 = -20 \text{ m/s}^2$

$$v_{xf} = 40 - 5(t + \Delta t)^2 = 40 - 5t^2 - 10t\Delta t - 5(\Delta t)^2$$

$$\Delta v_x = v_{xf} - v_{xi} = -10t\Delta t - 5(\Delta t)^2$$

$$a_x = \lim_{\Delta t \to 0} \frac{\Delta v_x}{\Delta t} = \lim_{\Delta t \to 0} (-10t - 5\Delta t) = -10t$$

$$a_x = (-10)(2.0) \text{ m/s}^2 = \boxed{-20 \text{ m/s}^2}$$

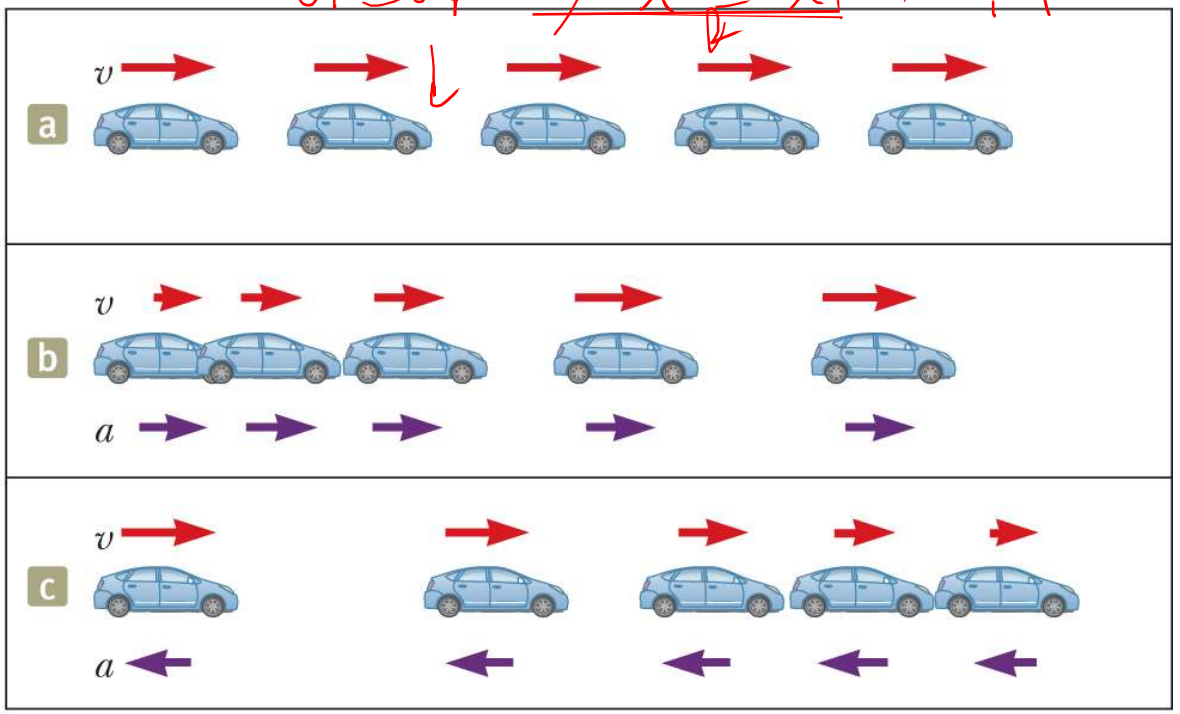


# Motion Diagrams

This car moves at constant velocity (zero acceleration).

This car has a constant acceleration in the direction of its velocity.

This car has a constant acceleration in the direction opposite its velocity.



# Quick Quiz 2.6

Which one of the following statements is true?

(a) If a car is traveling eastward, its acceleration must be eastward.

(b) If a car is slowing down, its acceleration must be negative.

(c) A particle with constant acceleration can never stop and stay stopped.

# Quick Quiz 2.6

Which one of the following statements is true?

(a) If a car is traveling eastward, its acceleration must be eastward.

(b) If a car is slowing down, its acceleration must be negative.

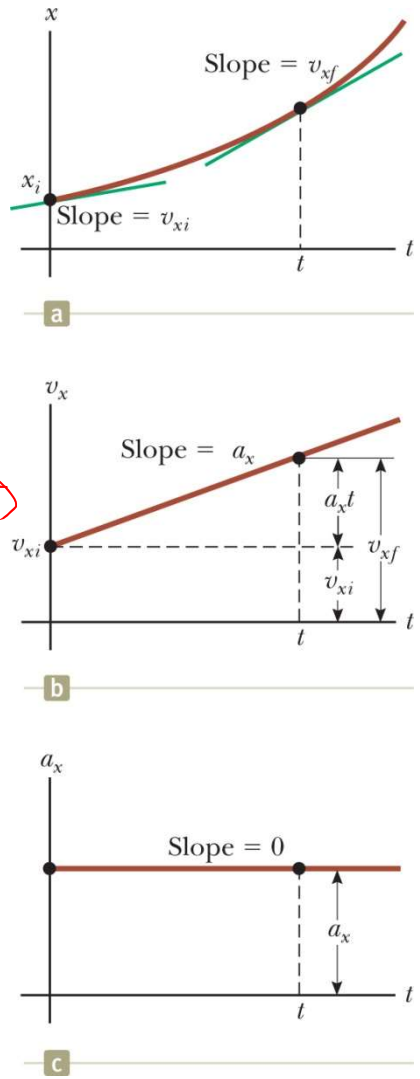
**(c) A particle with constant acceleration can never stop and stay stopped.**

# Analysis Model: Particle Under Constant Acceleration

$$a_x = \frac{v_{xf} - v_{xi}}{t - 0}$$

$$v_{xf} = v_{xi} + a_x t \quad (\text{for constant } a_x)$$

$$v_{x,\text{avg}} = \frac{v_{xi} + v_{xf}}{2} \quad (\text{for constant } a_x)$$



# Analysis Model: Particle Under Constant Acceleration

$$v_{x,\text{avg}} = \frac{\Delta x}{\Delta t}, \quad v_{x,\text{avg}} = \frac{v_{xi} + v_{xf}}{2}$$

$$\Delta x = x_f - x_i, \quad \Delta t = t_f - t_i = t - 0 = t$$

$$x_f - x_i = v_{x,\text{avg}} t = \frac{1}{2}\left(v_{xi} + v_{xf}\right)t$$

$$x_f = x_i + \frac{1}{2}\left(v_{xi} + v_{xf}\right)t \quad \left(\text{for constant } a_x\right)$$

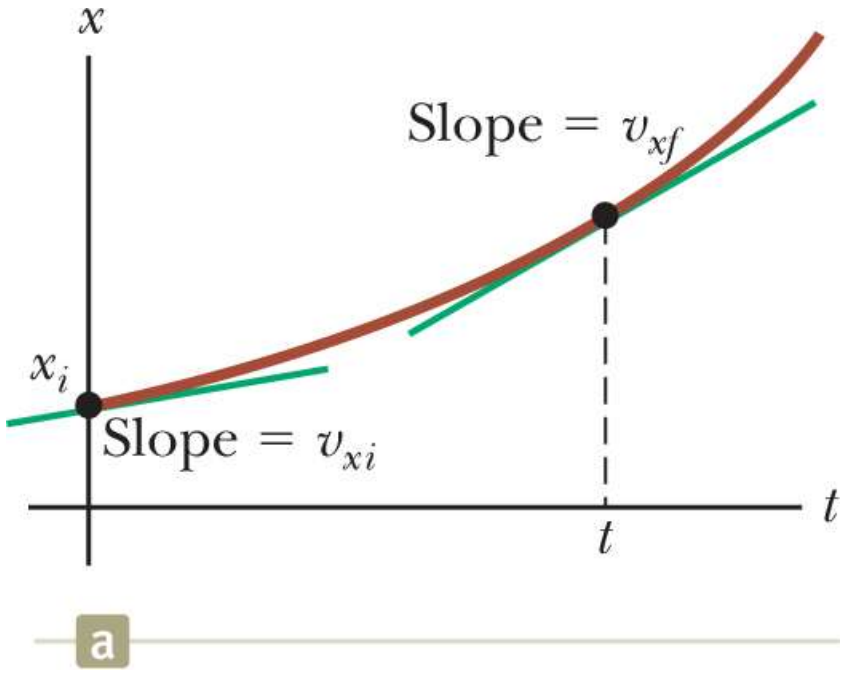
# Analysis Model: Particle Under Constant Acceleration

$$v_{xf} = v_{xi} + a_x t \quad \text{substitute into } x_f = x_i + \frac{1}{2}\left(v_{xi} + v_{xf}\right)t$$

$$x_f = x_i + \frac{1}{2}\left[v_{xi} + \left(v_{xi} + a_x t\right)\right]t$$

$$x_f = x_i + v_{xi}t + \frac{1}{2}a_x t^2$$

$$\left(\text{for constant } a_x\right)$$



# Analysis Model: Particle Under Constant Acceleration

$$v_{xf} = v_{xi} + a_x t \rightarrow t = \frac{v_{xf} - v_{xi}}{a_x}$$

substitute into $x_f = x_i + \frac{1}{2}\left(v_{xi} + v_{xf}\right)t$

$$x_f = x_i + \frac{1}{2}\left(v_{xi} + v_{xf}\right)\left(\frac{v_{xf} - v_{xi}}{a_x}\right) = x_i + \frac{v_{xf}^2 - v_{xi}^2}{2ax}$$

$$v_{xf}^2 = v_{xi}^2 + 2a_x\left(x_f - x_i\right) \quad \left(\text{for constant } a_x\right)$$

# Analysis Model: Particle Under Constant Acceleration

$$v_{xf} = v_{xi} + a_x t \qquad x_f = x_i + v_x t + \frac{1}{2} a_x t^2$$

$$\left.\begin{aligned} v_{xf} &= v_{xi} = v_x \\ x_f &= x_i + v_x t \end{aligned}\right\} \text{ when } a_x = 0$$

# Kinematic Equations

$$v_{xf} = v_{xi} + a_x t \qquad (2.13)$$

$$v_{x,\text{avg}} = \frac{v_{xi} + v_{xf}}{2} \qquad (2.14)$$

$$x_f = x_i + \frac{1}{2}\left(v_{xi} + v_{xf}\right)t \qquad (2.16)$$

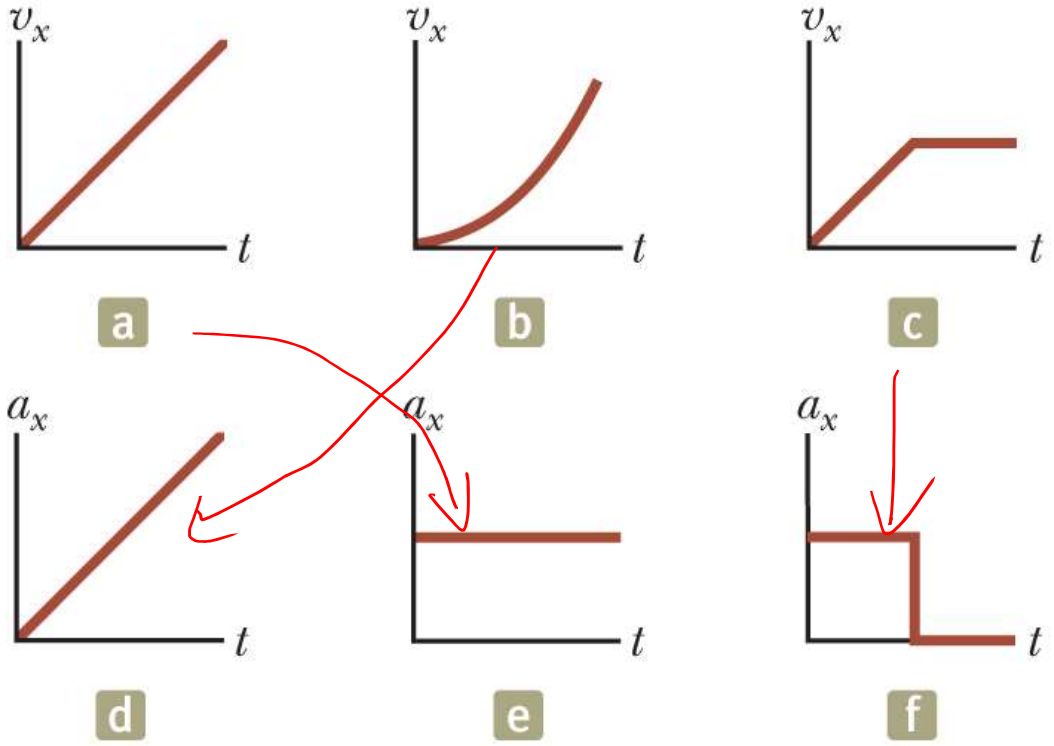
$$x_f = x_i + v_{xi}t + \frac{1}{2}a_x t^2 \qquad (2.15)$$

$$v_{xf}^{\,2} = v_{xi}^{\,2} + 2a_x\left(x_f - x_i\right) \qquad (2.17)$$

(Constant acceleration only!)

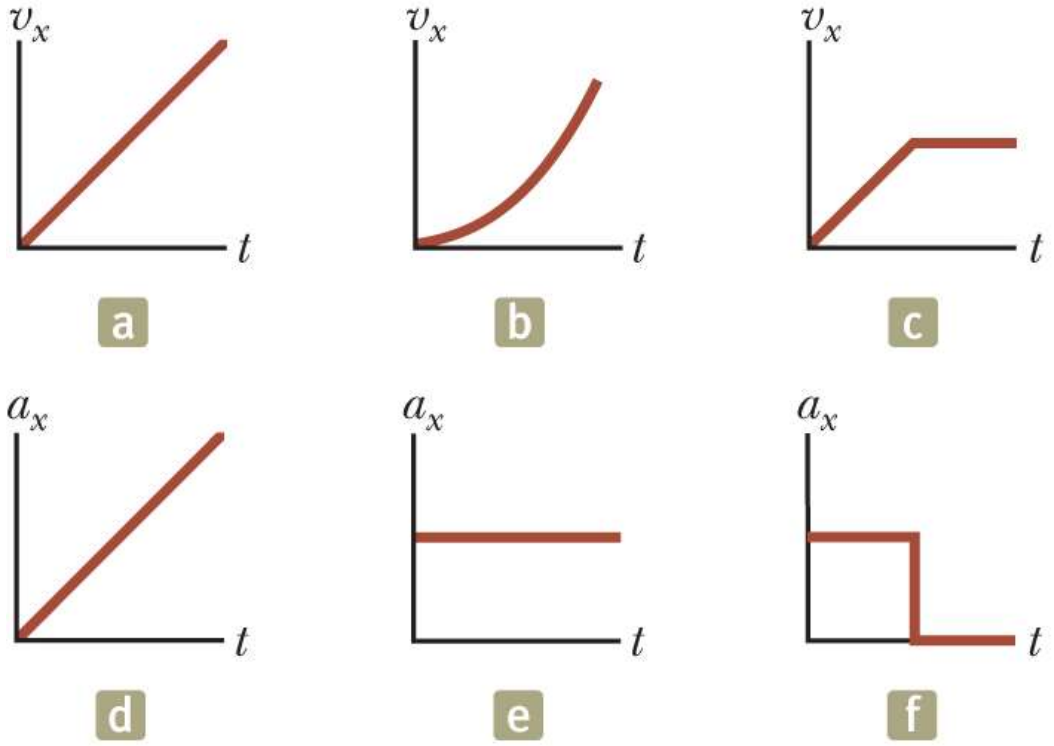
# Quick Quiz 2.7

In the figure, match each $v_x$–$t$ graph on the top with the $a_x$–$t$ graph on the bottom that best describes the motion.



# Quick Quiz 2.7

In the figure, match each $v_x$–$t$ graph on the top with the $a_x$–$t$ graph on the bottom that best describes the motion.




(a)–(e), (b)–(d), (c)–(f)

# Analysis Model: Particle Under Constant Acceleration

$$v_{xf} = v_{xi} + a_x t$$

$$v_{x,\text{avg}} = \frac{v_{xi} + v_{xf}}{2}$$

$$x_f = x_i + \frac{1}{2}\left(v_{xi} + v_{xf}\right)t$$

$$x_f = x_i + v_{xi}t + \frac{1}{2}a_x t^2$$

$$v_{xf}^{\,2} = v_{xi}^{\,2} + 2a_x\left(x_f - x_i\right)$$



# Example 2.7:
# Carrier Landing

A jet lands on an aircraft carrier at a speed of 140 mi/h ( ≈ 63 m/s).

(A) What is its acceleration (assumed constant) if it stops in 2.0 s due to an arresting cable that snags the jet and brings it to a stop?

$$a_x = \frac{v_{xf} = v_{xi}}{t} \approx \frac{0 - 63 \text{ m/s}}{2.0 \text{ s}} = \boxed{-32 \text{ m/s}^2}$$

# Example 2.7: Carrier Landing

(B) If the jet touches down at position $x_i = 0$, what is its final position?

$$x_f = x_i + \frac{1}{2}\left(v_{xi} + v_{xf}\right)t$$

$$= 0 + \frac{1}{2}\left(63\ \text{m/s} + 0\right)\left(2.0\ \text{s}\right) = \boxed{63\ \text{m}}$$

# Example 2.7: Carrier Landing

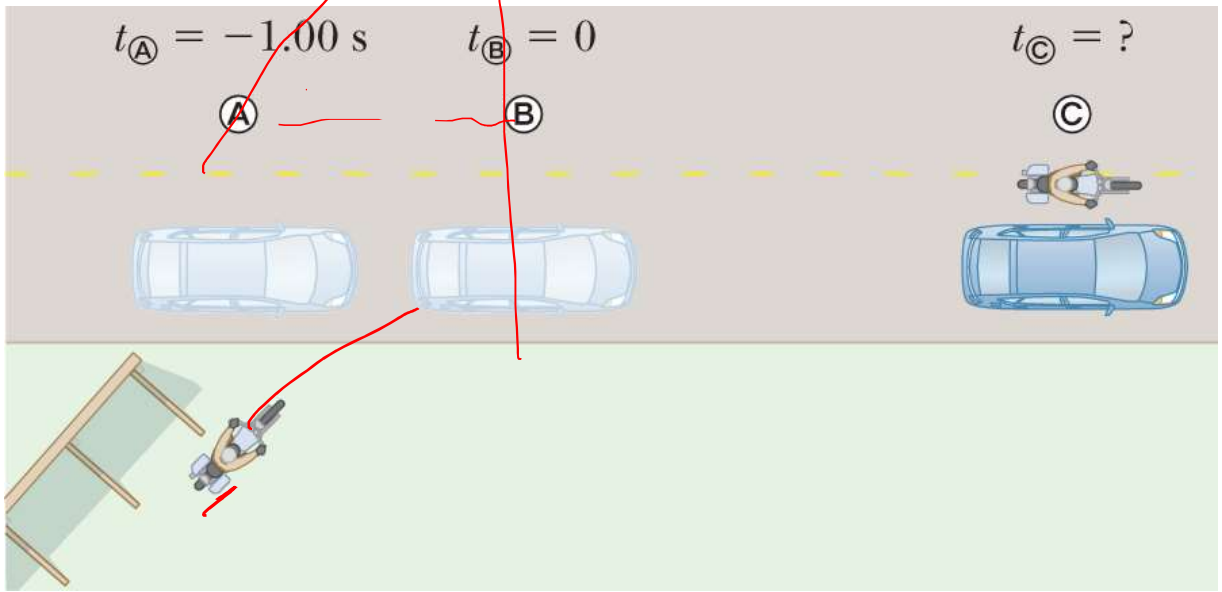
Suppose the jet lands on the deck of the aircraft carrier with a speed faster than 63 m/s but has the same acceleration due to the cable as that calculated in part (A). How will that change the answer to part (B)?

The position coordinate $x_f$ will be larger.

$$x_f = x_i + \frac{1}{2}\left(v_{xi} + v_{xf}\right)t$$

# Example 2.8:
# Watch Out for the Speed Limit!

You are driving at a constant speed of 45.0 m/s when you pass a trooper on a motorcycle hidden behind a billboard. One second after your car passes the billboard, the trooper sets out from the billboard to catch you, accelerating at a constant rate of 3.00 m/s$^2$. How long does it take the trooper to overtake your car?



# Example 2.8:
# Watch Out for the Speed Limit!

$$x_{\text{car}} = x_{\text{B}} + v_{x\ \text{car}} t$$

$$x_f = x_i + v_{xi} t + \frac{1}{2} a_x t^2$$

$$x_{\text{trooper}} = 0 + (0)t + \frac{1}{2} a_x t^2 = \frac{1}{2} a_x t^2$$

$$x_{\text{trooper}} = x_{\text{car}}$$

$$\frac{1}{2} a_x t^2 = x_{\text{B}} + v_{x\ \text{car}} t$$

# Example 2.8:
# Watch Out for the Speed Limit!

$$\frac{1}{2}a_x t^2 - v_{x\ \text{car}}\, t - x_B = 0$$

$$t = \frac{v_{x\ \text{car}} \pm \sqrt{v_{x\ \text{car}}^2 + 2a_x x_B}}{a_x} = \frac{v_{x\ \text{car}}}{a_x} \pm \sqrt{\frac{v_{x\ \text{car}}^2}{a_x{}^2} + \frac{2x_B}{a_x}}$$

$$t = \frac{45.0\ \text{m/s}}{3.00\ \text{m/s}^2} + \sqrt{\frac{(45.0\ \text{m/s})^2}{(3.00\ \text{m/s}^2)^2} + \frac{2(45.0\ \text{m})}{3.00\ \text{m/s}^2}} = \boxed{31.0\ \text{s}}$$

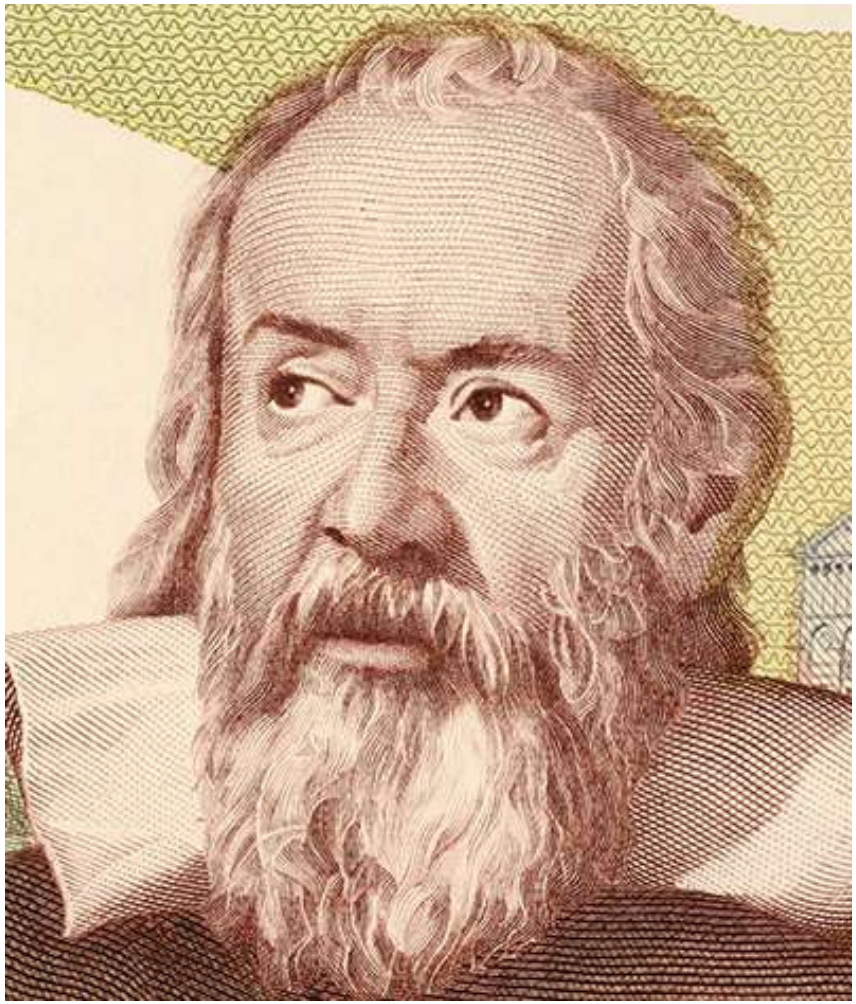
# Example 2.8:
# Watch Out for the Speed Limit!

What if the trooper had a more powerful motorcycle with a larger acceleration? How would that change the time at which the trooper catches your car?

$$t < 31 \text{ s}$$

$$t = \frac{v_{x\ \text{car}}}{a_x} + \sqrt{\frac{v_{x\ \text{car}}^2}{a_x^{\ 2}} + \frac{2x_{\text{B}}}{a_x}}$$

# Freely Falling Objects



Galileo Galilei



Leaning Tower of Pisa

# Quick Quiz 2.8 Part I

What happens to the acceleration of a ball after it is thrown upward into the air? (Neglect air resistance.)

(a) increases

(b) decreases

(c) increases and then decreases

(d) decreases and then increases

(e) remains the same

# Quick Quiz 2.8 Part I

What happens to the acceleration of a ball after it is thrown upward into the air? (Neglect air resistance.)

(a) increases

(b) decreases

(c) increases and then decreases

(d) decreases and then increases

**(e) remains the same**

# Quick Quiz 2.8 Part II

What happens to the speed of a ball after it is thrown upward into the air? (Neglect air resistance.)

(a) increases

(b) decreases

(c) increases and then decreases

(d) decreases and then increases

(e) remains the same

# Quick Quiz 2.8 Part II

What happens to the speed of a ball after it is thrown upward into the air? (Neglect air resistance.)

(a) increases

(b) decreases

(c) increases and then decreases

**(d) decreases and then increases**

(e) remains the same

# Conceptual Example 2.9: The Daring Skydivers

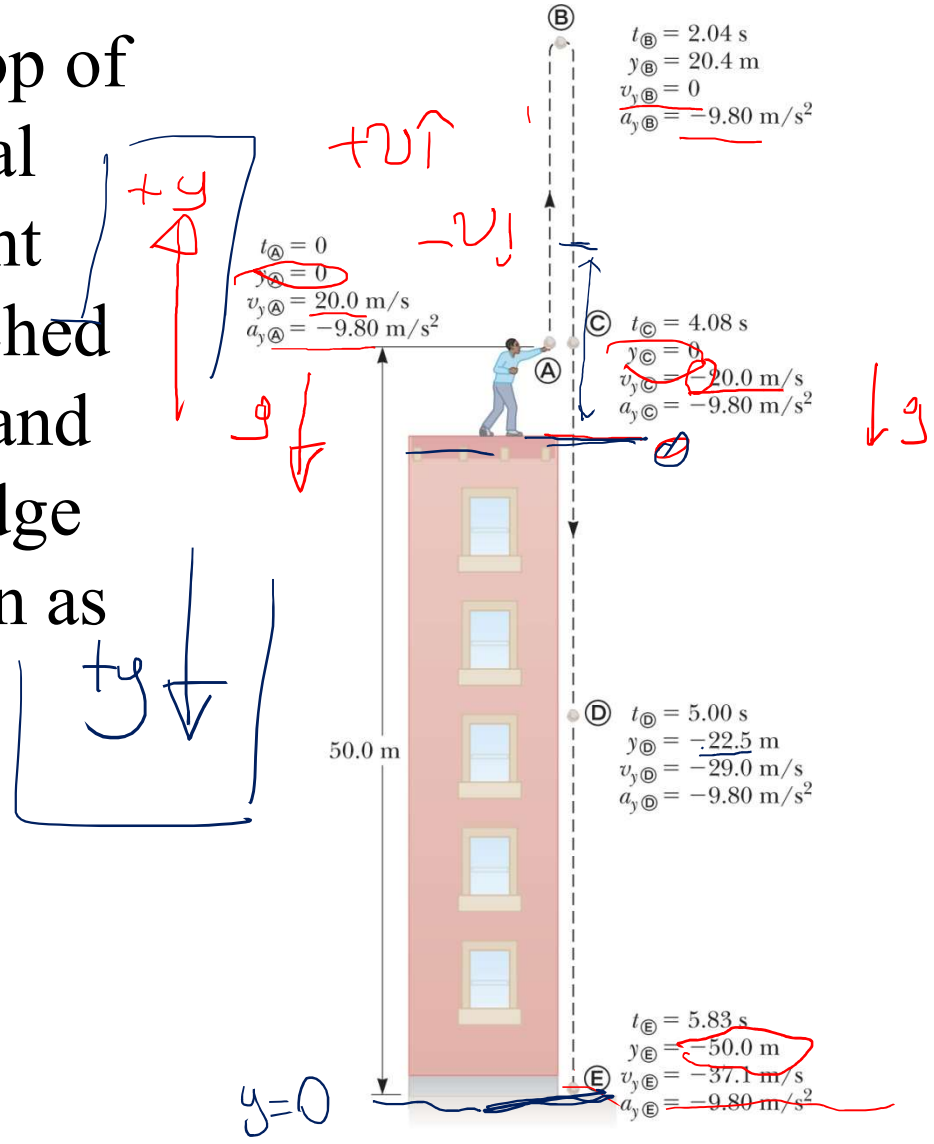
A skydiver jumps out of a hovering helicopter. A few seconds later, another skydiver jumps out, and they both fall along the same vertical line. Ignore air resistance so that both skydivers fall with the same acceleration. Does the difference in their speeds stay the same throughout the fall? Does the vertical distance between them stay the same throughout the fall?

**The difference in their speeds remains the same throughout the fall.**

**The first skydiver covers a greater distance than the second → the separation distance between them increases.**

# Example 2.10:
# Not a Bad Throw for a Rookie!

A stone thrown from the top of a building is given an initial velocity of 20.0 m/s straight upward. The stone is launched 50.0 m above the ground, and the stone just misses the edge of the roof on its way down as shown in the figure.



# Example 2.10:
# Not a Bad Throw for a Rookie!

(A) Using $t_A = 0$ as the time the stone leaves the thrower's hand at position $A$, determine the time at which the stone reaches its maximum height.

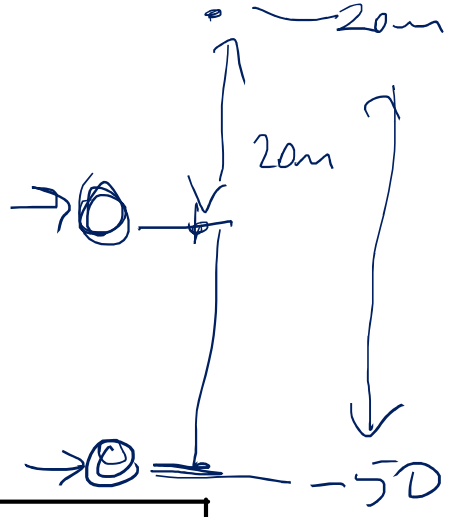
$$v_{yf} = v_{yi} + a_y t \Rightarrow t = \frac{v_{yf} - v_{yi}}{a_y} = \frac{v_{yB} - v_{yA}}{-g}$$

$$t = t_B = \frac{0 - 20.0 \text{ m/s}}{-9.80 \text{ m/s}^2} = \boxed{2.04 \text{ s}}$$

# Example 2.10:
# Not a Bad Throw for a Rookie!

(B) Find the maximum height of the stone (above its initial position).

$$y_{\max} = y_B = y_A + v_{xA}t + \frac{1}{2}a_y t^2$$




$$y_B = 0 + (20.0\ \text{m/s})(2.04\ \text{s}) + \frac{1}{2}(-9.8\ \text{m/s}^2)(2.04\ \text{s})^2 = \boxed{20.4\ \text{m}}$$

# Example 2.10:
# Not a Bad Throw for a Rookie!

(C) Determine the velocity of the stone when it returns to the height from which it was thrown.

$$v_{yC}^{\,2} = v_{yA}^{\,2} + 2a_y\left(y_C - y_A\right)$$

$$v_{yC}^{\,2} = \left(20.0\ \text{m/s}\right)^2 + 2\left(-9.80\ \text{m/s}^2\right)\left(0-0\right)$$

$$= 400\ \text{m}^2/\text{s}^2$$

$$v_{yC} = \boxed{-20.0\ \text{m/s}}$$

# Example 2.10:
# Not a Bad Throw for a Rookie!

(D) Find the velocity and position of the stone at $t = 5.00$ s.

$$v_{yD} = v_{yA} + a_y t$$

$$= 20.0 \text{ m/s} + \left(-9.80 \text{ m/s}^2\right)\left(5.00 \text{ s}\right) = \boxed{-29.0 \text{ m/s}}$$

$$y_D = y_A + v_{yA} t + \frac{1}{2} a_y t^2$$

$$= 0 + \left(20.0 \text{ m/s}\right)\left(5.00 \text{ s}\right) + \frac{1}{2}\left(-9.80 \text{ m/s}^2\right)\left(5.00 \text{ s}\right)^2$$

$$= \boxed{-22.5 \text{ m}}$$

# Example 2.10:
# Not a Bad Throw for a Rookie!

What if the throw were from 30.0 m above the ground instead of 50.0 m? Which answers in parts (A) to (D) would change?

**None of the answers would change.**

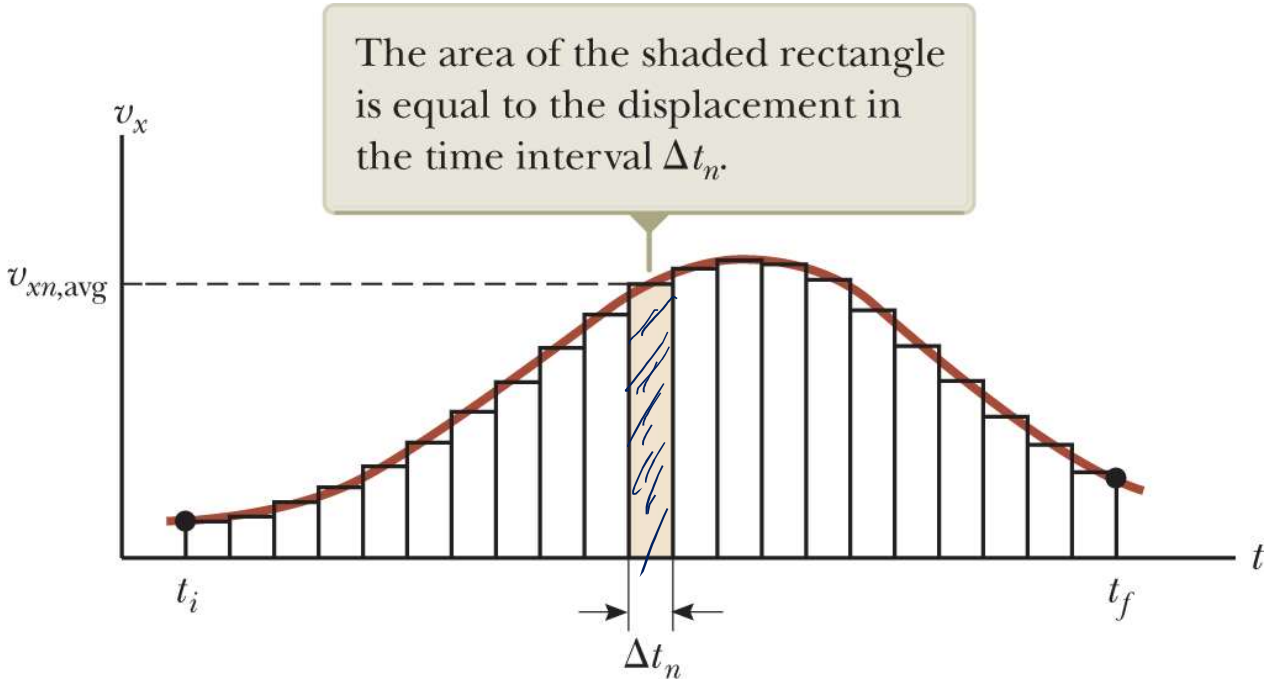
# Kinematic Equations Derived from Calculus

$$\Delta x_n = v_{xn,\text{avg}} \Delta t_n \qquad \Delta x = \sum_n v_{xn,\text{avg}} \Delta t_n \qquad \Delta x = \lim_{\Delta t_n \to 0} \sum_n v_{xn,\text{avg}} \Delta t_n$$

Riemann Sum

$v(t) = at + bt^2$

$$\Delta x = \int_{t_i}^{t_f} v_x(t)\, dt$$



# Kinematic Equations Derived from Calculus

$$a_x = \frac{dv_x}{dt} \Rightarrow dv_x = a_x dt$$

$$v_{xf} - v_{xi} = \int_0^t a_x dt$$

$$v_{xf} - v_{xi} = a_x \int_0^t dt = a_x (t - 0) = a_x t$$

$$v_{xf} = v_{xi} + a_x t$$

# Kinematic Equations Derived from Calculus

$$v_x = \frac{dx}{dt} \Rightarrow \int_{x_i}^{x_f} dx = \int_0^t v_x\,dt$$

$$\boxed{x_f - x_i = \int_0^t v_x\,dt}$$

$$v_x = v_{xf} = v_{xi} + a_x t$$

$$x_f - x_i = \int_0^t (v_{xi} + a_x t)\,dt$$

$$= \int_0^t v_{xi}\,dt + a_x \int_0^t t\,dt = v_{xi}(t-0) + a_x\left(\frac{t^2}{2} - 0\right)$$

$$x_f - x_i = v_{xi}t + \frac{1}{2}a_x t^2$$

# Assessing to Learn

Michael is going to the store 6 miles away. He rides his bike at 12 mph for the first half of the trip, then walks at 4 mph for the remainder.
Michael's average speed for the trip to the store is closest to:

1. 2 mph 2. 4 mph 3. 6 mph
4. 8 mph 5. 10 mph 6. 12 mph
7. 14 mph
8. exactly halfway between two of the values above
9. impossible to determine

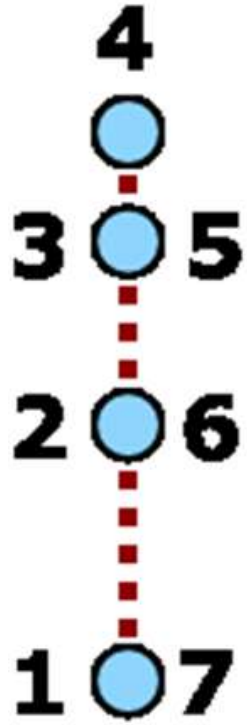
# Assessing to Learn

Wendy walks 10 m in one direction at 2 m/s, then runs 6 m in the same direction at 6 m/s. Next, she stops for 4 seconds, and finally walks in the opposite direction at 4 m/s for 6 seconds.
Wendy's average speed is **closest** to:

1. ½ m/s 2. 1 m/s 3. 1½ m/s
4. 2 m/s 5. 2½ m/s 6. 3 m/s
7. 3½ m/s 8. 4 m/s
9. the negative of one of the choices above
10. impossible to determine

# Assessing to Learn

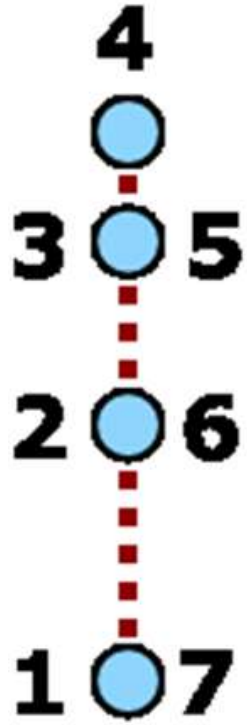
A ball is thrown straight up into the air. Its position at 7 instants of time are shown in the figure; the maximum height is reached at position 4. At which of the labeled points is the **speed** of the ball **smallest**?




1. point 1 2. point 2 3. point 3
4. point 4 5. point 5 6. point 6
7. point 7
8. exactly 2 of the points shown
9. more than 2 of the points shown
10. impossible to determine

# Assessing to Learn

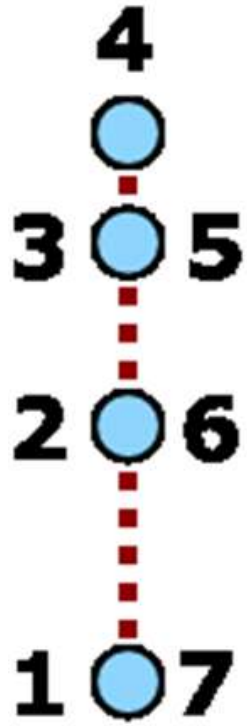
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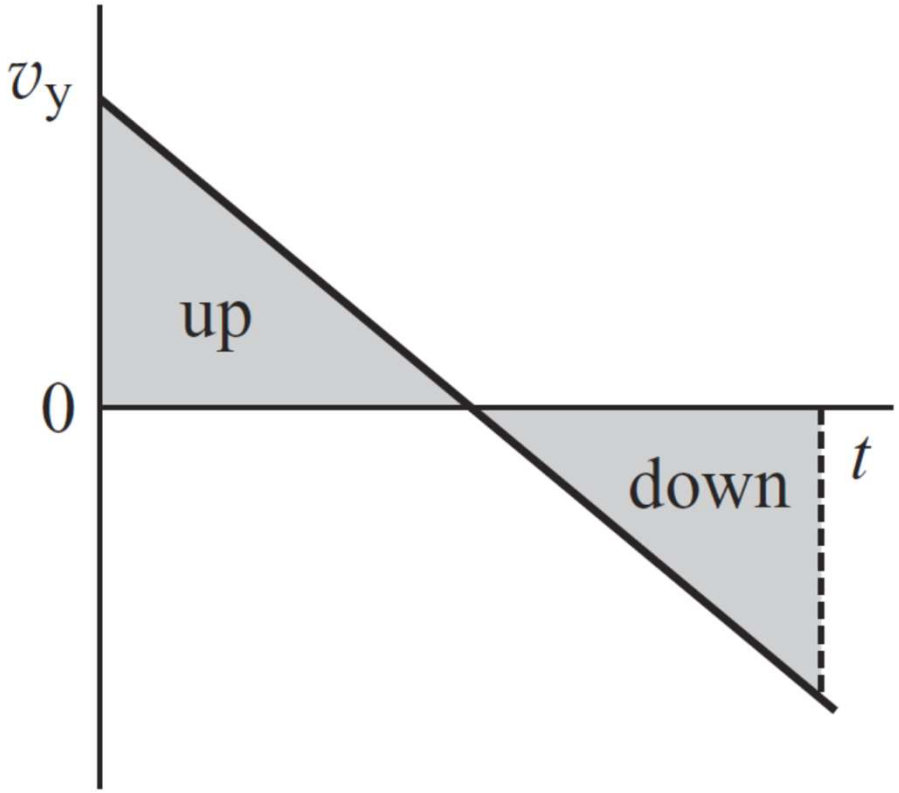
# Assessing to Learn

A ball is thrown straight up into the air. Its position at 7 instants of time are shown in the figure; the maximum height is reached at position 4. At which of the labeled points is the **speed** of the ball **largest**?



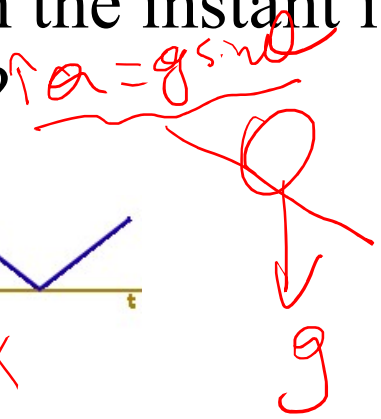

1. point 1 2. point 2 3. point 3
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8. exactly 2 of the points shown
9. more than 2 of the points shown
10. impossible to determine

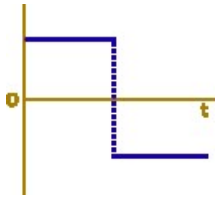
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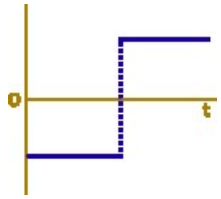


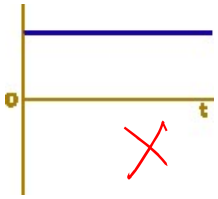
# Assessing to Learn

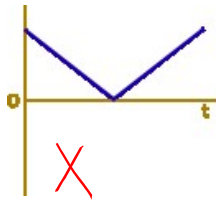
A ball is rolled up an incline so that it goes part-way up and then rolls back down. Which of the graphs below could represent its acceleration vs. time from the instant it is released until it returns to where it started?

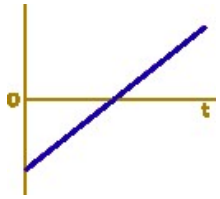


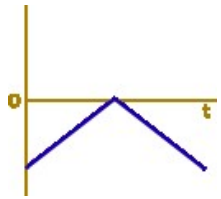

1. 

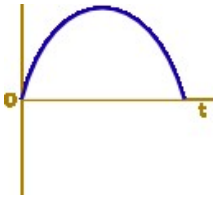

2. 

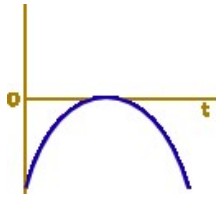

3. 


4. 


5. 


6. 

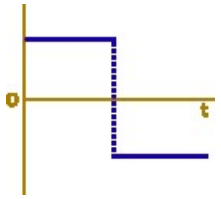

7. 

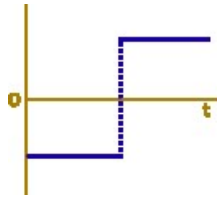

8. 

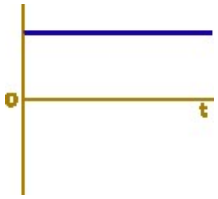

9. None of the graphs  10. Two or more of the graphs

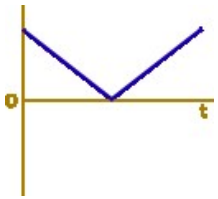
# Assessing to Learn

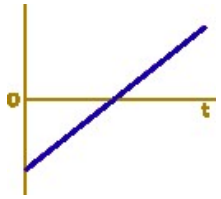
A ball is rolled up an incline so that it goes part-way up and then rolls back down. Which of the graphs below could represent its velocity vs. time from the instant it is released until it returns to where it started?

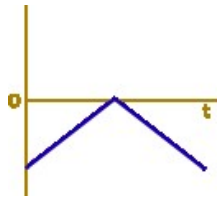
1. 

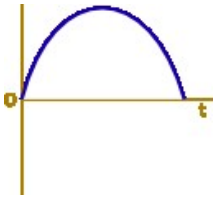

2. 

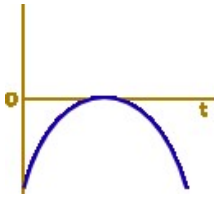

3. 


4. 


5. 


6. 


7. 


8. 


9. None of the graphs  10. Two or more of the graphs

# Assessing to Learn

Ralph walks 3 m to the left in 2 seconds, then 4 m to the right in 3 seconds. Next, he stops for 3 seconds, and finally walks 5 m to the left in 4 seconds.
Ralph's final position is **closest** to:

1. 0 m 2. 2 m 3. 4 m 4. 6 m
5. 8 m 6. 10 m 7. 12 m 8. 14 m
9. The negative of one of the choices above
10. Impossible to determine

# Assessing to Learn

Ralph walks 3 m to the left in 2 seconds, then 4 m to the right in 3 seconds. Next, he stops for 3 seconds, and finally walks 5 m to the left in 4 seconds.
The distance Ralph travels is **closest** to:

1. 0 m 2. 2 m 3. 4 m 4. 6 m
5. 8 m 6. 10 m 7. 12 m 8. 14 m
9. The negative of one of the choices above
10. Impossible to determine

# Assessing to Learn

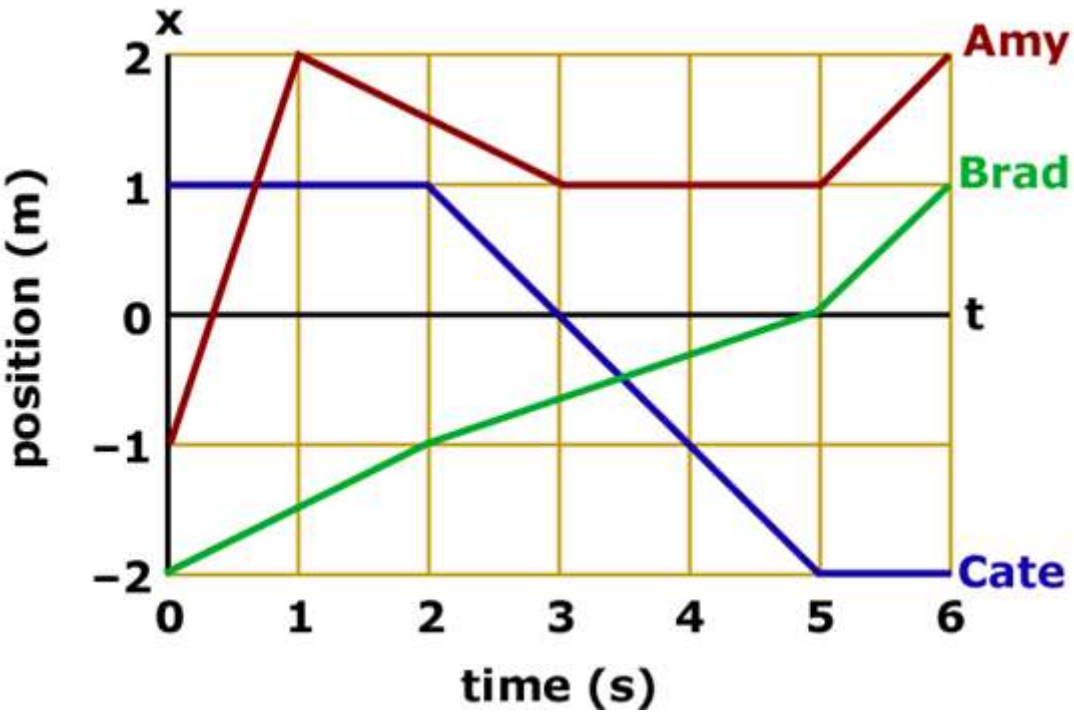
Ralph walks 3 m to the left in 2 seconds, then 4 m to the right in 3 seconds. Next, he stops for 3 seconds, and finally walks 5 m to the left in 4 seconds. Ralph's displacement is **closest** to:

1. 0 m 2. 2 m 3. 4 m 4. 6 m
5. 8 m 6. 10 m 7. 12 m 8. 14 m
9. The negative of one of the choices above
10. Impossible to determine

# Assessing to Learn

Amy, Brad, and Cate are walking (or running) along a straight line as represented below.
Which people have the same displacement?

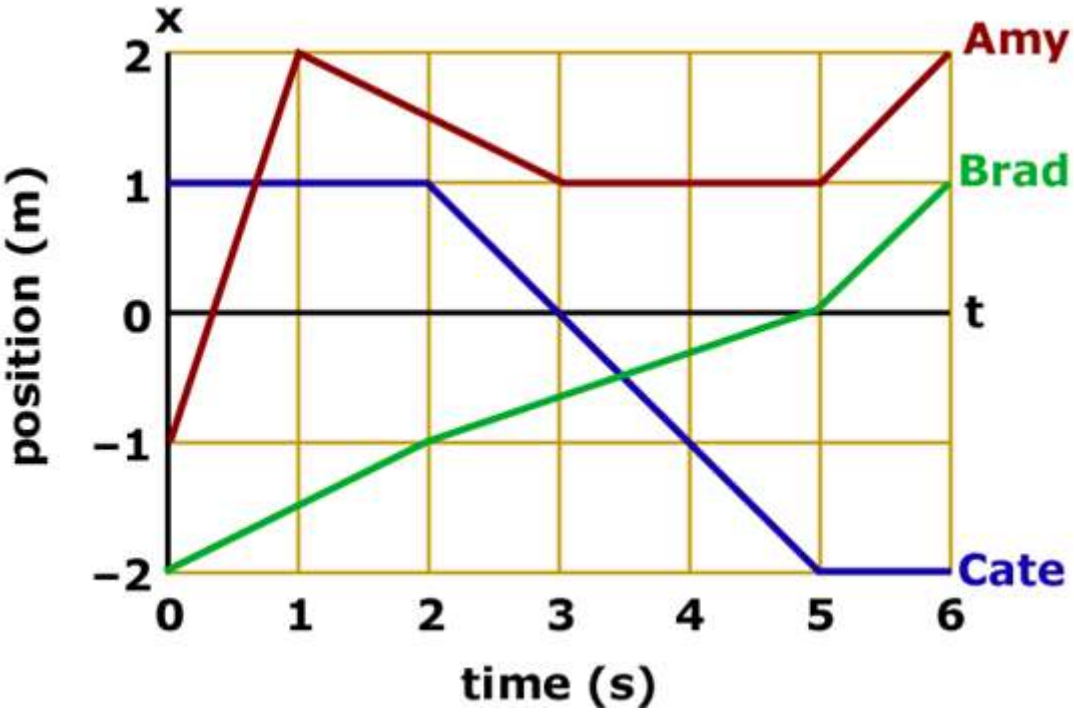
1. none; they all have different displacements
2. Amy and Brad
3. Amy and Cate
4. Brad and Cate
5. all three are the same



# Assessing to Learn

Amy, Brad, and Cate are walking (or running) along a straight line as represented below.
Which people have the largest displacement?

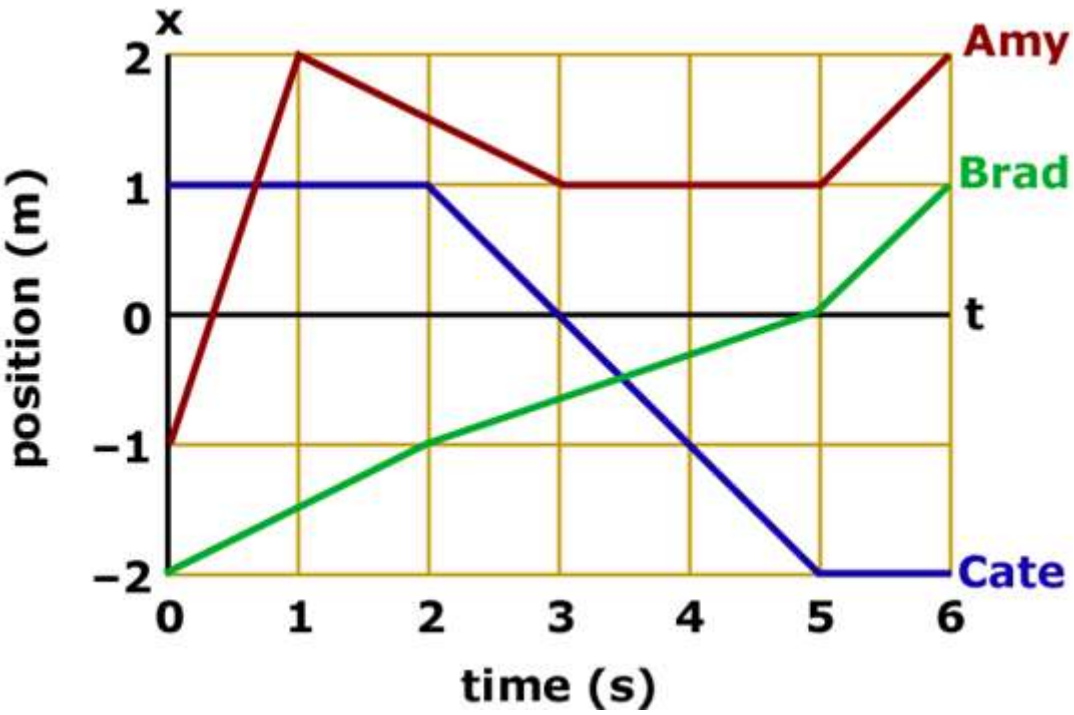
1. only Amy
2. only Brad
3. only Cate
4. Amy and Brad
5. Amy and Cate
6. Brad and Cate
7. all are the same
8. impossible to determi



# Assessing to Learn

Amy, Brad, and Cate are walking (or running) along a straight line as represented below.
Amy's distance traveled is closest to:

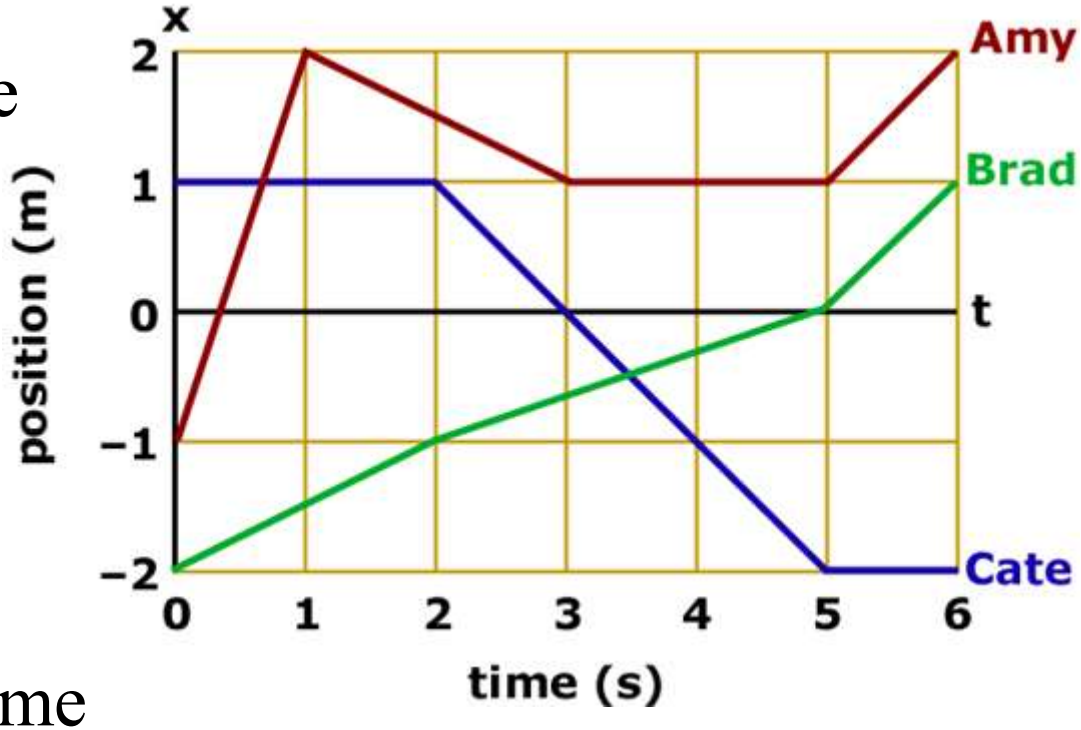
1. 1 m
2. 3 m
3. 5 m
4. 7 m
5. 9 m
6. 11 m
7. 13 m
8. 15 m
9. 17 m
10. exactly halfway between two of the values above



# Assessing to Learn

Amy, Brad, and Cate are walking (or running) along a straight line as represented below.
Which people have the same average speed during the time period shown?

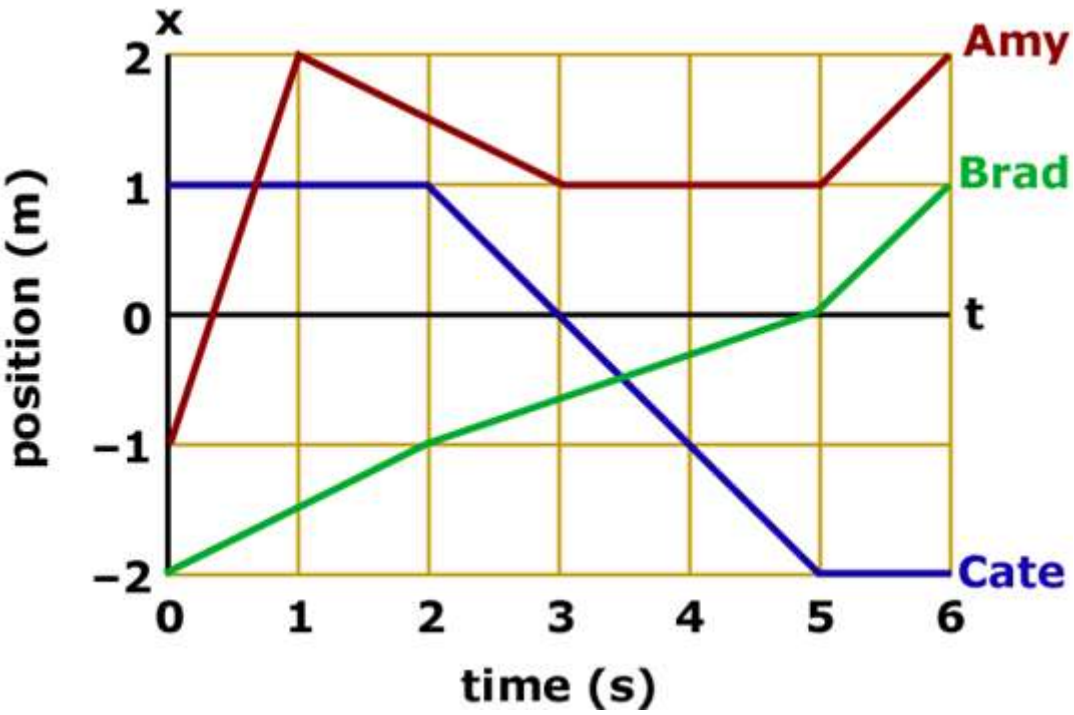
1. none; they all have different average speeds
2. Amy and Brad
3. Amy and Cate
4. Brad and Cate
5. all three are the same



# Assessing to Learn

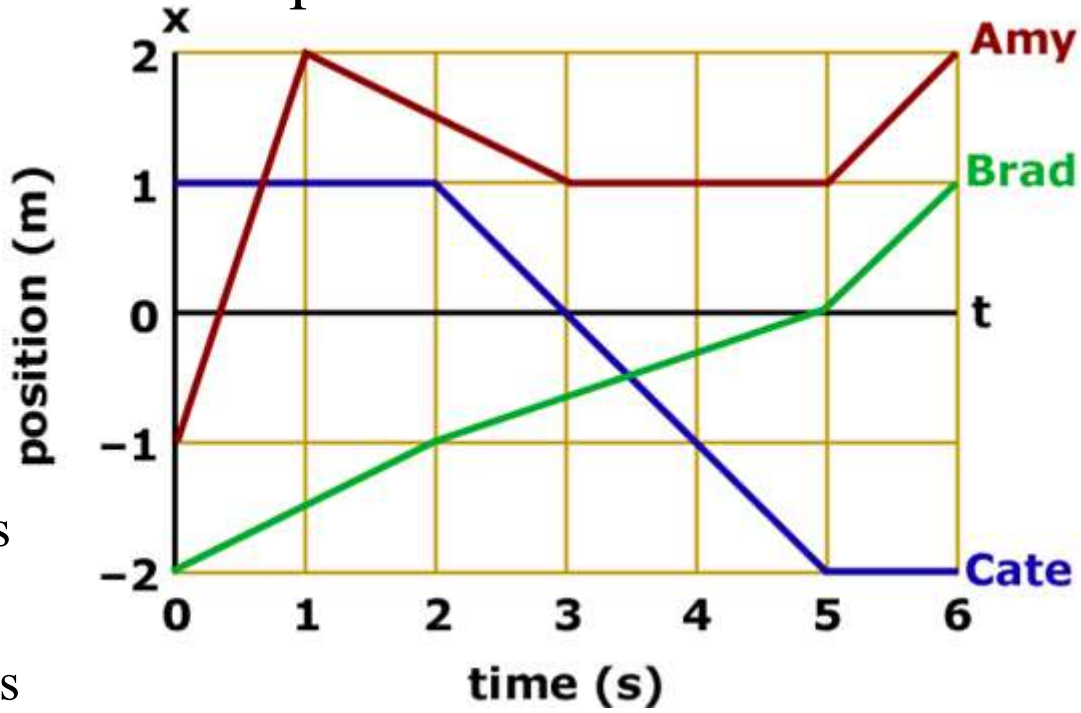
Amy, Brad, and Cate are walking (or running) along a straight line as represented below.
Which people are moving toward the origin at $t = 2\frac{1}{2}$ s?

1. None
2. Amy only
3. Brad only
4. Cate only
5. Amy and Brad
6. Amy and Cate
7. Brad and Cate
8. All three



# Assessing to Learn

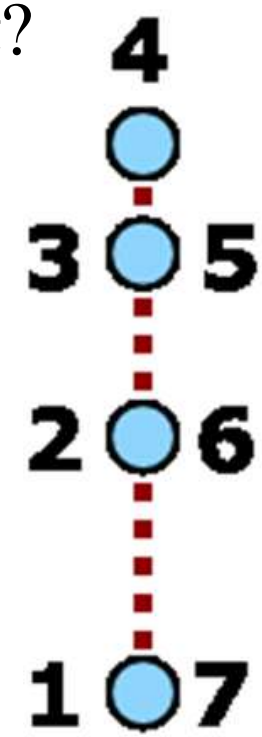
Amy, Brad, and Cate are walking (or running) along a straight line as represented below.
During which 1-second time period(s) are there at least two people with the same speed?




1. 0–1 s
2. 1–2 s
3. 2–3 s
4. 3–4 s
5. 4–5 s
6. 5–6 s
7. None of the time intervals
8. Two of the time intervals
9. Three of the time intervals
10. Four or more of the time intervals

# Assessing to Learn

A ball is thrown straight up into the air. Its position at 7 instants of time are shown in the figure; the maximum height is reached at position 4. At which of the labeled points is the **acceleration** of the ball **smallest**?




1. point 1 2. point 2 3. point 3
4. point 4 5. point 5 6. point 6
7. point 7
8. exactly 2 of the points shown
9. more than 2 of the points shown
10. impossible to determine